

Short-term electricity load forecasting—A systematic approach from system level to secondary substations

Marco G. Pinheiro ^{a,b,*}, Sara C. Madeira ^c, Alexandre P. Francisco ^d

^a Instituto Superior Técnico, Universidade de Lisboa, Portugal

^b EDP, Portugal

^c LASIGE, Faculdade de Ciências, Universidade de Lisboa, Portugal

^d INESC-ID Lisboa, Portugal

ARTICLE INFO

Keywords:

Load forecasting
System level
Secondary substations
Ensemble model
Generalized additive models
Interpretable models

ABSTRACT

Energy forecasting covers a wide range of prediction problems in the utility industry, such as forecasting demand, generation, price, and power load over time horizons and different power levels. Short-term load forecasting allows the system operator to make important decisions during network management and planning, which represents an economic improvement in the global electrical system, as well as acts as a fundamental component to address challenges imposed by the energy transition. Despite the urgent need, research into the current state-of-the-art in low voltage forecasting, other than at the smart meters level, has not yet been carried out in an in-depth way. We propose a systematic approach from system level to low voltage considering not only the performance of the models but also the applicability, interpretability, and reproducibility of the method/model. An initial benchmark model was compared against improved regression models enhanced by introducing new synthetic explanatory variables and a different regression technique, generalized additive models. The error was reduced by 42%–47% compared to the benchmark model, preserving the interpretability. Additionally, a simple ensemble method was evaluated to determine how it improves accuracy for specific periods, such as weekends, summer vacation, or public holidays, in which modeling is particularly difficult using standalone generalized additive models. The method was applied to the national power load and, for the first time, to all 100,000 secondary substations that integrate the Portugal power grid, rather than to tackle the few open datasets in this area. The approach was used to build a daily forecasting system *live* called PREDIS (Portuguese acronym for Distributed Prediction) whose outcomes are used to anticipate load peaks and network constraints in the context of the Portugal distribution system operator.

1. Introduction

1.1. Motivation

Since the first days of electricity distribution, utilities forecast the electricity demand for the next hours, days, and months. At that time, when electricity was essentially used for public illumination, forecasting demand was as easy as counting the number of light bulbs and multiplying by the power they consume. Nowadays, energy forecasting is much more complex. Electricity is consumed promptly almost anywhere and at any time by pressing a switch. Moreover, the energy transition, as the cornerstone of the upcoming developments of the energy system, includes the complex interaction of multiple technologies, business models innovation, and the decline of established business models and technologies [1]. Challenges arise with upcoming increasing complexity, connectivity, scale, and scope:

- Ability to accommodate more green energy towards a 100% renewable energy system. As renewables grow, through the deployment of wind and solar energy, the availability of power has become less dispatchable. Wind and solar energy cannot be turned on and off at will, they are intermittent, come and go depending on the weather and time of day, so they must be accommodated [2]. On the other hand, wind power also suffers from curtailment due to grid constraints or abundance of wind power during low demand for electricity.
- Ability to meet the increasing charging needs of electric vehicles (EV), which represents a large amount of electrification in terms of the demand for electricity in the following years. EVs are considered an important factor in the decarbonization of a large part of transport, and the power system must be able to feed them with

* Corresponding author at: Instituto Superior Técnico, Universidade de Lisboa, Portugal.

E-mail addresses: marco.pinho@tecnico.ulisboa.pt (M.G. Pinheiro), sacmadeira@fc.ul.pt (S.C. Madeira), aplf@tecnico.ulisboa.pt (A.P. Francisco).

<https://doi.org/10.1016/j.apenergy.2022.120493>

Received 10 September 2022; Received in revised form 20 November 2022; Accepted 3 December 2022

Available online 19 December 2022

0306-2619/© 2022 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

renewable energy. This integration could be helped by managing smart charging technology according to local grid restrictions and supply abundance, preferably renewable generation. This would provide greater flexibility to the network and prevent investments in new electrical infrastructure [3]. Furthermore, the battery on wheels has been considered an asset rather than a problem, in vehicle-to-grid (V2G), vehicle-to-home (V2H), and other usage and business models.

- Ability to manage dispatchable demand as more assets and models are introduced to increase flexibility on the demand side. When the power grid is overloaded or when the generation mix is not so green, controlled demand could be reduced instantly to lighten the system load. Demand-side management aggregators, virtual power plants, and logical dispatch of decentralized energy resources (DER) are models to add flexibility to the grid [2].
- Ability to store heating and electricity during low demand and high (renewable) energy availability, through centralized or decentralized technologies such as capacitors, superconductors, flywheels, batteries, heating storage, compressed air, pumped hydro, and (green) hydrogen (or other power-to-gas) [2,4]. The diverse storage technology ecosystem presents different characteristics in terms of the technology readiness level (TRL), the readiness of the business model, the storage capacity, the duration of storage, and the velocity or readiness for discharge.

These challenges need to be tackled from various perspectives, including energy forecasting as a fundamental enabler. Not only at the system level, but also at primary and secondary substations and at each energy point of delivery or generation as well. The benefits of new controllable loads or the storage of surplus electricity production are easily understandable given the challenges faced [3] and also leveraging those challenges in precise and exhaustive load forecasting. Energy forecasting remains a research topic to be covered.

As the system becomes increasingly complex and at the center of the energy transition, various business needs of energy forecasting have been reinforced within utilities [5], such as non-dispatchable generation forecasting [6], demand forecasting [7], trading in the gross market [8], transmission and distribution planning [9], operations and maintenance [10], demand side management [11], decentralized energy resources (DER) optimization, and financial planning. It is applicable across the whole value chain: production, transmission, distribution, retail, and trading. As an important tool, the accuracy of the forecast translates into the financial performance of energy utilities. A conservative estimate is that a 1% reduction in forecast error for a 10 GW utility can save up to \$ 1.6 million annually [12]. Another estimate is that a 0.1% improvement in forecasting in a midsize European utility can help reduce about \$ 3 million in operating costs in imbalance markets [13] considering the ability to forecast every meter, transformer feeder, beyond forecasting at the system level, and consequently enabling dynamic tariffs, demand response, and DER management to shave peak demand.

1.2. Literature survey

There is no single forecast that can meet all utility needs. Different forecasts and approaches are used depending on the purpose. Electricity forecasting can be classified by three aspects:

- the object which is being estimated, such as wind generation, solar generation, wave energy generation, hydroelectric energy generation, load forecasting, spatial load forecasting, demand forecasting, and energy price forecasting;
- the time horizon [5], which defines how far in advance the model forecasts, classified into four categories: very short-term – VSTF – that spans up to 1 h, short-term – STF – that spans up to 1 or 2 weeks, medium-term – MTF – that spans up to 1 or 3 years, and long-term – LTF – that spans up to 30 years; although the definitions of the time horizon differ by author, it determines both the updating cycle and the relevance of the explanatory variables;

- the aggregation level, such as the international integrated power grid (like the European electrical network), the regional or national DSO power grid, the substation or secondary substation loads, the energy point of delivery (highly dependent on consumption type), and virtual power plant's or energy retail portfolios.

Table 1 associates business needs with the type of forecast and the time horizon.

In the context of the power grid, short-term load forecasting (STLF) is considered an important tool or input for maintenance, scheduling, planning of power generation (both centralized and distributed), load switching, safety evaluation, cost optimization, and the guarantee of continuous electricity supply in general [12,14,15].

A considerable number of comprehensive reviews have established the evidence in STLF for further research concerning the models and methods [16–26].

Regarding **models**, both machine learning-based models and classic forecasting models of time series based on mathematical and statistical modeling have been applied to model future load.

Some of the most widely used methods are the autoregressive techniques introduced by George Box and Gwilym Jenkins, such as the autoregressive moving average—**ARMA**. To use the ARMA model, an essential condition is that the time series should be stationary, which is achieved by differencing the non-stationary load time series in the first place [27]. A sequence that contains characteristics such as trend, seasonality, or periodicity is called a non-stationary sequence. Auto-regressive integrated moving average – **ARIMA** – is a well-known non-stationary time series model, which can reflect the changes of different data patterns, and the model requires fewer parameters to estimate [28]. To deal with seasonality, seasonal ARIMA – **SARIMA** was introduced. [29] is an example of applying SARIMA to predict the half-hour power load of the system one day in advance. Overall, these forecasting models are adaptable, can deal with seasonality and with non-stationary data, and only require the past value of a timeseries. However, it is unlikely to perform well on long-term predictions and they are unable to include exogenous variables that domain practitioners consider essential. [30,31] model nonlinear temperature effect using regression splines. Statistical methods, such as generalized linear models – **GLM** – [5,32] and generalized additive models – **GAM** – [33–35], have been used to model the relationship between variables, including exogenous explanatory variables. Probabilistic forecasting [22] has also been applied to STLF as partially linear additive quantile regression (PLAQR) [36], or kernel density estimation [37,38].

On the other hand, machine learning-based models have also been applied in STLF, such as artificial neural networks (ANN) [5,39–41], long short-term memory based (LSTM) [42,43], temporal fusion transformers-based (TFT) [44], support vector machines (SVM) [45, 46], fuzzy logic [47–49], random forest [50], particle swarm optimization (PSO) [51,52], genetic algorithms (GA) [43], and others that might combine more than one technique as meta-learning [53–55].

General public dynamics derived data from cellular network and the energy consumption dataset to predict the next 7 days of energy demand from a northern region of Italy using **random forest** [50]. Regarding **artificial neural networks**-based forecasters, most articles used the multilayer perceptron, usually a feed-forwarded fully connected or, alternatively, used a recurrent network. The activation function (either logistic or hyperbolic tangent functions), the number of hidden layers (mostly one or two), the number of output neurons (one output in a multi-step configuration – hour by hour forecasting – or multi-model – one different model for each hour –, or multivariate outputs – all 24-hours predicted at once), the number of hidden neurons (most papers chose this number by trial and error, finding the best forecasting accuracy without overfitting) are decisions to take during network design. A proposed framework based on **long short-term memory** (LSTM) recurrent neural networks for residential load forecasting performs better compared to various alternative state-of-the-art

Table 1
Type of forecasting in accordance with business requirements and time horizon.

	VST	ST	MT	LT	Object
Energy selling and purchasing					
Non-dispatchable generation	×	×	×		Generation
Electricity consumption	×	×	×		Demand
Trading Energy	×	×	×		Price
T&D planning			×	×	Load
Operation and maintenance	×	×			Load
Demand side management	×	×			Generation and Demand
Financial planning			×	×	Generation, Demand and Price

approaches [56] and concluded that the higher the inconsistency, the more LSTM can improve forecasting compared to simple backpropagation neural networks. **Fuzzy regression** considers that the deviations between the observed values and the estimated values are assumed to depend on the indefiniteness of the system structure, while in multiple linear regression the deviations are supposed to be errors in the observed values. There are examples that improve the precision of equivalent multiple linear regression [48]. **Genetic Programming (GP)** exploits the concept of evolution to tackle the search for possible model structures (or any computer program) and perform symbolic regression. It can be employed to search complex linear spaces. When selecting input variables, GP automatically finds the variables that contribute the most to the model and then constructs an equation [28]. Raza and Khosravi reviewed the characteristics, explanatory variables and importance of load forecasting, but mainly, they discussed the application of artificial neural networks as a superior performance over statistical techniques mainly when abrupt changes in environmental or socio-logical variables occur [57]. One interesting point of this reference, they explored the hybrid techniques that were proposed by published research with the combination of superior attributed of two or more algorithms. They concluded that the hybridization of two or more techniques shows better results for load forecast problem than one technique alone, either conventional statistical techniques or ANN. They refer to: (i) ANN with fuzzy and genetic algorithm, (ii) ANN with expert system and regression techniques, (iii) ANN with wavelet and time series, (iv) ANN with support vector machine and artificial immune system, (v) ANN with genetic algorithms, (vi) ANN with gradient-based learning techniques. **Meta-learning** based approaches have been also applied for load forecasting. Multiple meta-learners are used to rank the candidate load forecast models on their forecasting accuracy and, then, a scoring-voting mechanism weights recommendations from each meta-learner and make the final ensemble prediction [53]. **Transfer learning** and meta learning are adopted to transform general (one-for-all) models into high-performance individualized model customized using the specific individual's small dataset [54]. Error-correction can also be used to learn with the prediction loss on recent observations [55].

Nevertheless, as [22] points out, most studies pursue the goal of finding the best technique for load forecasting, resulting in rather worthless articles. Either because virtually most papers focus on showing the superiority of the introduced technique on very specific datasets hiding their weaknesses, or lack of detailed information on the setup experiments or over-manipulating the data (for instance, excluding days whose electricity load are more unpredictable) [26]. *A universally best technique simply does not exist*, it is the data and the business needs that determine which technique is more useful.

Apart from focusing on the best technique, many articles also demonstrate how a methodology is used to solve the load forecasting problem or its sub-problems. General **methodological approaches** were identified into six general categories and comprehensively reviewed after with the fundamental benefits and drawbacks (based on [22,24,58]). Table 2 outlines the six general categories and techniques.

1.3. Knowledge gap and main contributions

Although a diverse set of load forecasting techniques and methodologies has been studied, there are still knowledge gaps on the subject of low voltage (LV) load forecasting [59] as follows:

- Load forecasting has been extensively applied either at the system/region level or at the building/point-of-delivery scope. Short-term load forecasting at the low voltage level, other than at the smart meter level, such as primary and secondary substations, has not been as extensive [59]. However, this does not mean that efforts have not been made. This research and Refs. [60,61] should be considered.
- Instead of evaluating the developed method only for performance (accuracy), practitioners should consider the impact and applicability of the method, as well, toward a live system implemented and adopted by the energy sector player.
- LV load forecasts can benefit from weather forecasts, particularly when these predictions are derived from multiple weather stations or from numerical weather predictions (NWP), whose significant advances have not yet been translated into improved LV load forecasts.
- In the context of LV load forecasting, numerous articles did not use benchmarks to compare their models beyond benchmark persistence forecast.
- Due to the scarcity of LV timeseries (Irish CER dataset and UK Low Carbon London trial data), one line of study would be to compare models using multiple datasets from different sources, rather than tackle only the few open datasets.
- There is a need for developing pragmatic methods to achieve the explainability of AI and ML methods in the LV load forecasting context, as well as make the models computationally cheap in order to tackle thousands of low voltage assets and predict its load curve for the next few days in a useful time to business needs.

This study seeks to contribute with an approach specifically tailored to STLF at the secondary substation level, an area of research that has been sparse. With that goal in mind, the following steps are considered in depth detail: (i) evaluation criteria considering the applicability, interpretability, reproducibility and accuracy aspects, (ii) benchmark against a classical regression model for system-level forecasting described by Tao Hong and his research group [5], and (iii) modeling input variables associated with a consistent description of parameter tuning and alternatives from both the statistical and the energy domain point of view.

Using this strategy, it will be possible to apply STLF to a large number of different time series at secondary substation level – because the parameter tuning and method details chosen at the system level might change when applied to the others, a consistent description will keep the reproducibility –, keep a communication bridge between load forecasters, operators, planning manager, and executives when considering aspects more than accuracy, such as interpretability and applicability – note how the load forecaster developed needs to be approved by managers, understandable by operators, and defensible before the regulator –, and the analysis of the results when this systematic methodology is applied, for the first time, to the all 100,000 secondary

Table 2
Short-term load forecasting methods based on [22,24,58].

Method	Description
Similar Pattern • Similar Day • Pattern Sequence • Sequence Learning	Determines the load curve as a sequence of various similar load profiles.
Variable Selection • Stepwise Method • Correlation • Mutual Information • Filtering • Optimization Algorithm • Time-dependent	Presumes the load curve behaves like a series of variables either correlated or independent from each other.
Hierarchical Forecasting • Bottom-up • Top-down • Ensemble • Weight Combination	Considers the data as an aggregated load, which is highly varying by changes in the load at lower levels of hierarchy.
Weather Station Selection • Average Model • Optimal-number-of-stations Model	Determines the best-fitted weather data into the load model.
Decomposition • Wavelet Transform • Empirical/Variational/Dynamic Mode Decomposition • Singular Spectrum Analysis • Double-layer Decomposition	Decompose and extract the characteristics of load time series before modeling them.
Meta-Learning • Stacking • Transfer Learning • Error correction	Learn from the output or model structure of other machine learning algorithms that have already been trained on data.

substations which integrates the Portugal power grid. Fig. 1 diagrams the scope of the study across the voltage levels of the power grid. In particular, the methodology is as follows:

- Using identical explanatory variables, we compare a GAM-based regression model with a classical one. Accuracy is assessed using distinct metrics while keeping in mind that a biased evaluation could result from choosing only one metric. This would serve as a benchmark base for LV load forecasting;
- Develop a method to improve the GAM-based regression model by introducing new synthetic explanatory variables based on the same data. New variables are introduced based on domain knowledge and a systematic approach. Numerical weather predictions were used as explanatory variables in the low voltage context;
- Compare the improved GAM-based regressor to a simple ensemble method, where weaker forecasters (also GAM-based) improve accuracy while still maintaining desirable interpretability;
- Use of a new dataset that encompasses all 100,000 secondary substations of the Portuguese electricity grid, for example, the grid level where energy is converted from medium voltage to low voltage using power transformers, rather than tackle the few open datasets.
- The methods and algorithms were implemented and adopted by Portugal's DSO as a live system capable of handling thousands of load curve predictions in a useful time and enabler of innovation (consider the impact of this research area)
- Describe the underdeveloped aspects in the context of LV load forecasting.

The reader might ask, given the huge efforts of the scientific community on the subject over several years, why their techniques have been mostly evaluated otherwise than against the secondary substation's time series. The point is that most Distributed System Operators (DSO) have just rolled out smart meters in secondary substations in the last three to six years [62–64] plus the time it takes to get a proper

period of data. Furthermore, no or partial subsets of these data have been made public. In contrast, this study addresses the entire dataset that has been collected from secondary substations in Portugal's power grid since 2015.

2. Material and methods

Two datasets (I and II) have been considered to conduct the study of six different models with different goals and techniques, which is summarized in Table 3.

2.1. Data description

Two datasets (I and II) have been considered to evaluate the proposed methodology and the forecasting model. Both datasets were not previously used to evaluate forecasters, following the recommendation to avoid overusing particular datasets [59]. Dataset I is related to the aggregated national power load, and, on the other hand, dataset II is the active energy measured on each of the 100,000 secondary substations. Since it includes all existing secondary substations in Portugal, dataset II is actually representative of the wider energy system. That will lessen the risk of modeling the power load based on spurious features and patterns of a biased dataset [59].

2.1.1. Dataset I

The national power load denotes the total electricity used by consumers or energy storage devices, plus the part that is naturally lost during energy transmission and distribution. In addition, it can be seen as the amount of energy produced in centralized power stations (hydro, solar, wind, natural gas, coal, oil, nuclear, etc.) added to the net electricity imported from neighboring countries. For the purpose of this chapter, the subject of analysis is the total load of the system minus the power used in pumped hydroelectric storage systems. The dataset

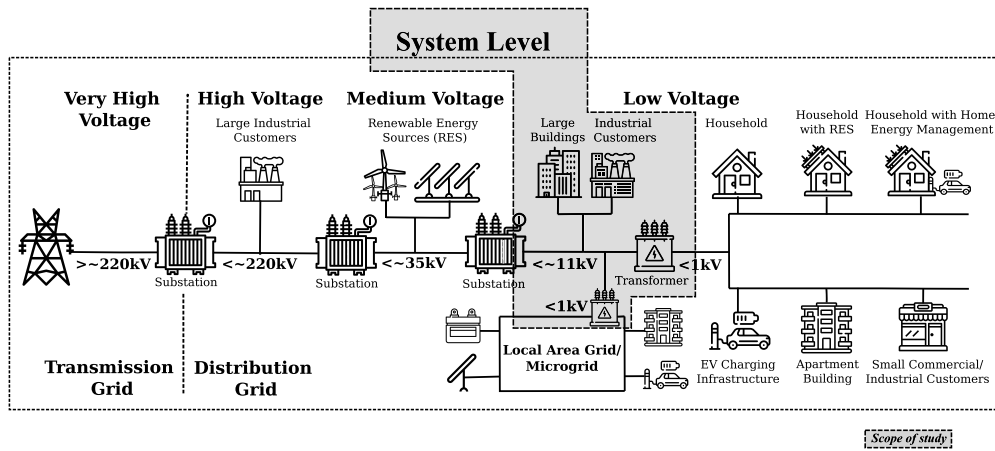


Fig. 1. The scope of the study includes the system level and the secondary substations of Portugal DSO grid including the private secondary substations of large electricity consumers at high and medium voltage.

Table 3

The study of six different models were conducted with two datasets depending on the model goal.

Section	Model	Goal	Dataset
2.2	GLMLF-B	System level GLM benchmark model	I
2.3	GAMLF-SL-M1	System level GAM model based on benchmark model	I
2.3.1	GAMLF-SL-M2	System level GAM model with lagged load covariates	I
2.3.2	GAMLF-SL-M3	System level GAM model with calendar covariates	I
2.4	GBMLF-SL	System level XGboost model	I
2.5	GAMLF-SLE	System level GAM and WMC-R ensemble model	I
2.6	GAMLF-SSL	Secondary substation level GAM model	II

is publicly available and can be downloaded from the TSO Data Hub website.¹

Formally, consider the dependent variable y as the national power load,

$$y = [y_1 \quad y_2 \quad \dots \quad y_t \quad \dots \quad y_T]^T. \quad (1)$$

Each entry y_t denotes the national power load (continental Portugal) in megawatt (MW) at time t which comprises 30 min period (down-sampled from original data).

The power load follows a pattern with annual, weekly, and daily seasonality. Fig. 2 shows the half-hour demand for a week from Monday to Sunday, in winter and summer. Note the weekly pattern: Working days are very similar, while Saturday and Sunday are different in both level and form. On working days, the three peaks are different depending on the season. Moreover, there are some important decreases in demand during the holidays of Christmas, the summer holidays, and the bank holidays.

Additionally, weather observations from 12 selected weather stations in Portugal are included in this dataset to study the effect of temperature on power load. The electricity load has a non-linear dependence on the temperature due to electrical heating and cooling, as Fig. 3 exhibits.

The weather data is publicly available and can be downloaded from the NOAA website.² The dataset includes meteorological observations such as temperature, dew point temperature, sea-level pressure, observable weather on a qualitative scale, and more. Data wrangling was performed to mitigate missing values, cope with temporal resolution differences between weather stations, and summarize the 12 weather station timeseries into one weighted average temperature as the “national temperature”. The weights reflect the 2013 annual electricity consumption of each region according to Table 4.

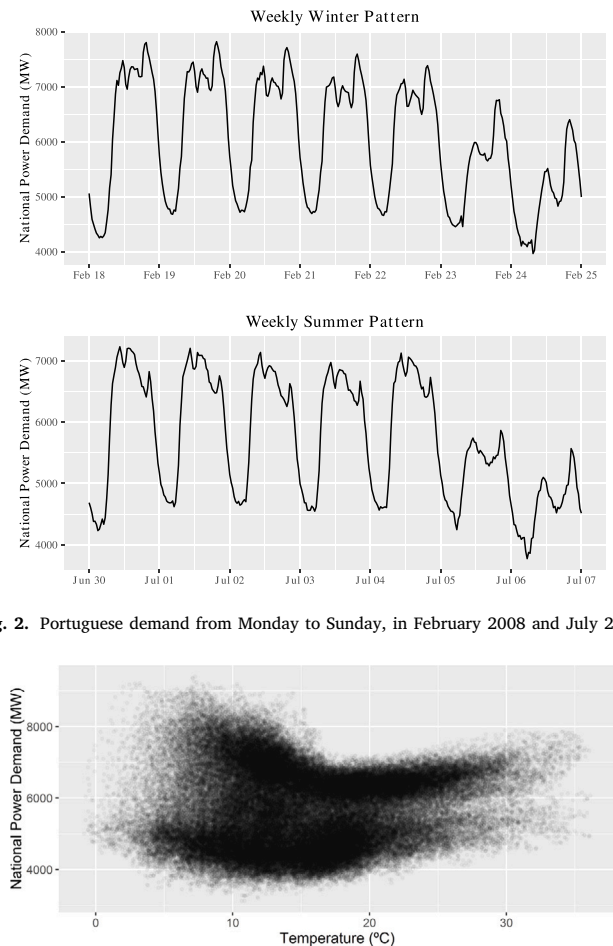


Fig. 2. Portuguese demand from Monday to Sunday, in February 2008 and July 2008.

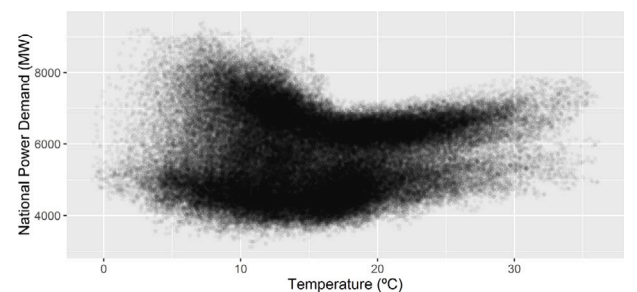


Fig. 3. Half-hourly national demand scattered plot with temperature observations.

¹ REN Data Hub website: <https://datahub.ren.pt/>.

² NOAA Data Access website: <http://www.ncdc.noaa.gov/data-access>.

Table 4

The weighted average of “national temperature” is computed from measurements of 12 selected weather stations whose weights are in accordance with the power demand of each region. In regions with more than one weather station selected, a simple average is calculated between them.

Region	Weight	Weather stations
Algarve	6.8%	Faro
Alentejo	7.8%	Beja and Portalegre
Lisboa	26.8%	Lisboa and Lisboa/Gago Coutinho
Centro	25.3%	Coimbra, Viseu, Castelo Branco, and Monte Real
Porto	15.2%	Porto
Norte	18.1%	Vila Real and Bragança

So the available data $\mathbf{x}_t$, which can be used to explain the dependent variable, includes trend index, calendar, time, and meteorological data as follows:

$$\mathbf{x}_t = \left[x_t^{(\text{Trend})} \quad x_t^{(\text{Month})} \quad x_t^{(\text{DayOfWeek})} \quad x_t^{(\text{TimeOfDay})} \quad x_t^{(\text{Temperature})} \quad x_t^{(\text{PublicHoliday})} \quad x_t^{(\text{DayOfYear})} \right]^T. \quad (2)$$

Let explain these components in more detail:

- $x_t^{(\text{Trend})}$ is a quantitative variable that represents the index for the entire range of available data. For example, 1 for the first half-hour of available data, 2 for the second half-hour, etc. This variable is useful for capturing the trend of increasing or decreasing power loads. The use of this variable might pose problems in model applicability due to the long trend that might change in significant events, for example, merging two utilities, recessions, economic booms, pandemics, and so on.
- $x_t^{(\text{Month})}$ is the index of the current month within the year from 1 to 12.
- $x_t^{(\text{DayOfWeek})}$ is a categorical variable representing the day of week—one category for each day: 1 for Sunday, 2 for Monday, 3 for Tuesday, 4 for Wednesday, 5 for Thursday, 6 for Friday, and 7 for Saturday.
- $x_t^{(\text{TimeOfDay})}$ is the index of the current time of day. Since this index represents each half-hour, its values range from 1 to 48 representing respectively the midnight and 11:30 p.m.
- $x_t^{(\text{Temperature})}$ is the meteorological covariate that represents the half-hourly “national temperature”—a weighted average computed from 12 weather stations in Portugal.
- $x_t^{(\text{PublicHoliday})}$ is a categorical variable representing the national holidays, including Carnival; a category for each public holiday. Some papers combine all public holidays into a unique boolean variable. This option may be more prudent due to the minority number of public holidays over one year. We have evaluated the two options, but the final model takes into account the most descriptive categorical variable.
- $x_t^{(\text{DayOfYear})}$ is a numerical variable representing the current day within the year. Its values range from 0 for 1st January and 1 for 31st December.

2.1.2. Dataset II

The disaggregated load denotes the power load in secondary substations, which is the interface between medium voltage (MV) and low voltage (LV) power system. Smart meters that track this local power load are usually installed close to power transformers that have the function to step down higher voltage to lower voltage. That conceptually splits the grid into different voltage levels, and in this case, the dataset II focuses on the challenging issue of the “last mile” of the distribution network. In contrast to primary substations that feed mainly a ring network scheme, in which more than one grid path is possible through grid maneuvers to prevent outages when a primary feeder faults, secondary substations feed mainly a radial network scheme, which distributed the energy to the final consumers as a “leaf” section of the grid. It is also important to note that the secondary substation can be owned by a unique client, and so the

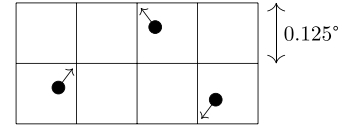


Fig. 4. Taking into account the NWP over the spatial grid box, each of the three substations (circles) is associated with the closest Euclidean-distant NWP.

electricity measurements are from a specific customer, usually a large building or an industrial customer. These secondary substations will be called PTC (client’s power transformer). Contrariwise, the secondary substations operated by the distribution system operator (DSO) are named PTD (distributor’s power transformer). For those, the electricity measurements refer to an aggregated set of consumers nearby, usually a neighborhood of a few dozens or even hundreds of low-voltage clients.

The data come from 96,989 secondary substations of Portugal mainland’s distribution grid, the entire system. It consists of 26,479 client’s (PTC) plus 70,510 DSO’s (PTD) secondary substations. The latter feeds 99.6% of electricity consumers (points of energy delivery) in Portugal, which, however, represents 47.4% of the energy consumed on the mainland of Portugal.

Formally the dependent variable y_n is the active power load (A^+) of the asset n in kilowatt (kW),

$$\mathbf{y}_n = [y_{1,n} \quad y_{2,n} \quad \dots \quad y_{t,n} \quad \dots \quad y_{T,n}]^T, \quad (3)$$

Numerical weather prediction (NWP) data is gathered with a 3 h time step over a spatial grid box, updated twice a day (00 and 12 UTC) for the next 72 h. However, the predictions are centrally available some hours late (between 7 and 8 h after the base time to which the prediction refers). The geographical discrete points are arranged in a two dimensional regular grid located every 0.125 degrees over longitude and latitude, in which it covers whole Portugal mainland.³ In this scenario, the challenge of assigning the most related weather station information to each secondary substation is minimized because there is a spatial grid box: Each secondary substation is associated with the closest euclidean-distant NWP, as shown in Fig. 4. Table 5 shows the numerical weather variables collected that are available for forecast purposes.

Furthermore, due to the different time resolution between the original load observations (15 min) and the meteorological predictions (3 h), the load observations are downsampled to 30 min using the average of its quarter-hour values, and the temperature is upsampled to 30 min using linear interpolation when the gap between two consecutive datapoints is not greater than 3 h. Missing values are removed from the dataset.

Formally, the matrix $\mathbf{X}_n$ are the calendar and meteorological variables that would be used as explanatory variables, where each entry $\mathbf{x}_{t,n}$ has a structure similar to that of Eq. (2),

$$\mathbf{x}_{t,n} = \left[x_t^{(\text{DayOfWeek})} \quad x_t^{(\text{PublicHoliday})} \quad x_t^{(\text{DayOfYear})} \quad x_t^{(\text{TimeOfDay})} \quad x_t^{(\text{Temperature})} \right]^T. \quad (4)$$

³ Latitude limits: $[36.5^\circ, 44.0^\circ]$; Longitude limits: $[-10.0^\circ, -5.5^\circ]$.

Table 5
The numerical weather prediction collected for modeling and inferring purposes.

Field	Symbol	Description	Unit
151	msl	Mean sea level pressure	Pa
167	2t	2 m temperature	K
169	ssrd	Surface solar radiation downwards (accumulated) ^a	J m ⁻²
228	tp	Total precipitation (accumulated) ^b	m
228 246	100u	100 m U wind component (towards east)	m s ⁻¹
228 247	100v	100 m V wind component (towards north)	m s ⁻¹

^aSSRD is accumulated over a particular period since the initial time step.

^bTotal precipitation is accumulated over a particular period since the initial time step. Units are the depth the water would have if it were spread evenly over the grid box.

Besides the components already introduced, $x_{t,n}^{(\text{Temperature})}$ is the last predicted temperature value for time t and the geographically closest point to asset n .

2.2. GLMLF-B

The generalized linear model-based load forecasting—benchmark model, GLMLF-B, is a classical regression model for system-level forecasting described by Tao Hong and his research group and it uses generalized linear regression [5].

2.3. GAMLF-SL

The generalized additive model-based load forecasting, GAMLF-SL, is a regression model for the system level. Generalized additive models (GAM) [65,66] involve a sum of smooth functions of covariates, thus capturing the non-linear effects of covariates on the dependent variable and maintaining an interpretable structure.

Considering the data available from dataset I, the modeling activity requires to define what is the best combination of available input variables, following the additive structure which GAM introduces. Therefore, the methodology carried forward has four steps: (i) formulation and selection of input variables (Eq. (2)), (ii) definition of model structure, (iii) model calibration and tuning, and (iv) evaluation of model and residuals. After residual analysis, the process is repeated to find a new set of input variables and model structure, focusing on the aspects or moments in which the residuals are higher. The process ends when a good balance is achieved between applicability, interpretability, reproducibility, and accuracy performance.

The initial structure of the model, GAMLF-SL-M1, was based on the GLMLF-B with the necessary modifications due to the new characteristics of the GAM technique. Namely, the qualitative variables *month* and *time of day* are changed to their quantitative versions, that is, the quantitative *day of year* and *time of day*⁴ variables, and smooth functions are used when appropriate. For instance, it is unnecessary to make explicit third-order polynomials for the temperature effect, since the smooth functions take care of this.

2.3.1. Lagged load

The power load timeseries, as many other signals, show the presence of repeating patterns or periodic signals. To understand the autocorrelation of timeseries, Figs. 5(a) and 5(b) exhibit the autocorrelation function (ACF) and the partial autocorrelation function (PACF). In both plots, high coefficients rise at immediately lag values and, more important, at 24 h lag, 2 days lag, 3 days lag, etc. This represents a daily seasonality. The amplitude of ACF of subsequently peaks decreases until 4 days lag but increases again until reaching a new maximum amplitude precisely at 7 days lag. This “scalped” shape is due to the weekly seasonality.

⁴ Here, the name (*time of day*) refers to the quantitative version. The interpretation is implicit from the context.

The autocorrelation functions suggest which are the best lag periods as autoregressive terms to be included in the model structure. However, the final selection of autoregressive terms depends on the applicability of the model. The use of a lagged power load as an input into the model leads to the necessity that the data should be available right on time to predict. For example, considering the 24-hour lagged power load as input, the model would predict a maximum horizon of 24 h, considering that there is no delay in the availability of the power load timeseries.

However, it is important to note that the covariates already included in the GAMLF-SL-M1 structure could capture these different seasonalities. To ensure that new autoregressive terms will improve the accuracy of the model despite additional complexity, ACF and PACF are applied to the residuals of GAMLF-SL-M1 instead of the power load signal. Figs. 5(c) and 5(d) suggest that even residuals manifest autocorrelation patterns and peaks are comparable with the ACF and PACF of the power load timeseries.

Taking into account the structure of the previous model, GAMLF-SL-M1, eleven different combinations of possible lagged power loads are added as candidate covariates and evaluated with a one-year update cycle. The best candidate add the covariates $f^{(\text{LagLoad24h})}(y_{t-48})$ and $f^{(\text{LagLoad1w})}(y_{t-336})$ to the new model GAMLF-SL-M2.

2.3.2. Calendar

The power load is largely dependent on human behavior, which in turn is conditioned by the calendar. The literature is full of various approaches to the calendar and time. For example:

- month (12 classes); months grouped by the four seasons (4 classes); months grouped into 7 types to distinguish the transitions between two adjacent seasons (7 classes); hot and cold days instead of months (2 classes);
- day of the week (7 classes); working and weekend days (2 classes); weekend, adjacent days to the weekend and other working days (3 classes); and other combinations of the 7 days;
- intraday steps (48 steps if half-hourly or 24 steps if hourly); or other combinations, for example: dawn, morning, lunch, afternoon, evening, and late night (6 classes);
- holidays treated separately; or treated as weekends; or even additionally modeling the surrounding days of a holiday apart from others.

The GAMLF-SL-M2 model already includes the 7 days of the week to discriminate the intraday power load into 7 typical curves, one per week. However, the residuals plotted over one year show patterns in specific days and periods. Fig. 6 shows the patterns in the summer holidays – August –, Christmas and New Year, and other surrounding public holidays.

Taking into account the structure of the previous model, GAMLF-SL-M2, six different adjustments are studied to cope with the existence of a holiday on the day the model forecasts, or the existence of a holiday on the day the autoregressive covariates uphold. After evaluating them with a one-year update cycle, the best candidate adjusts the number

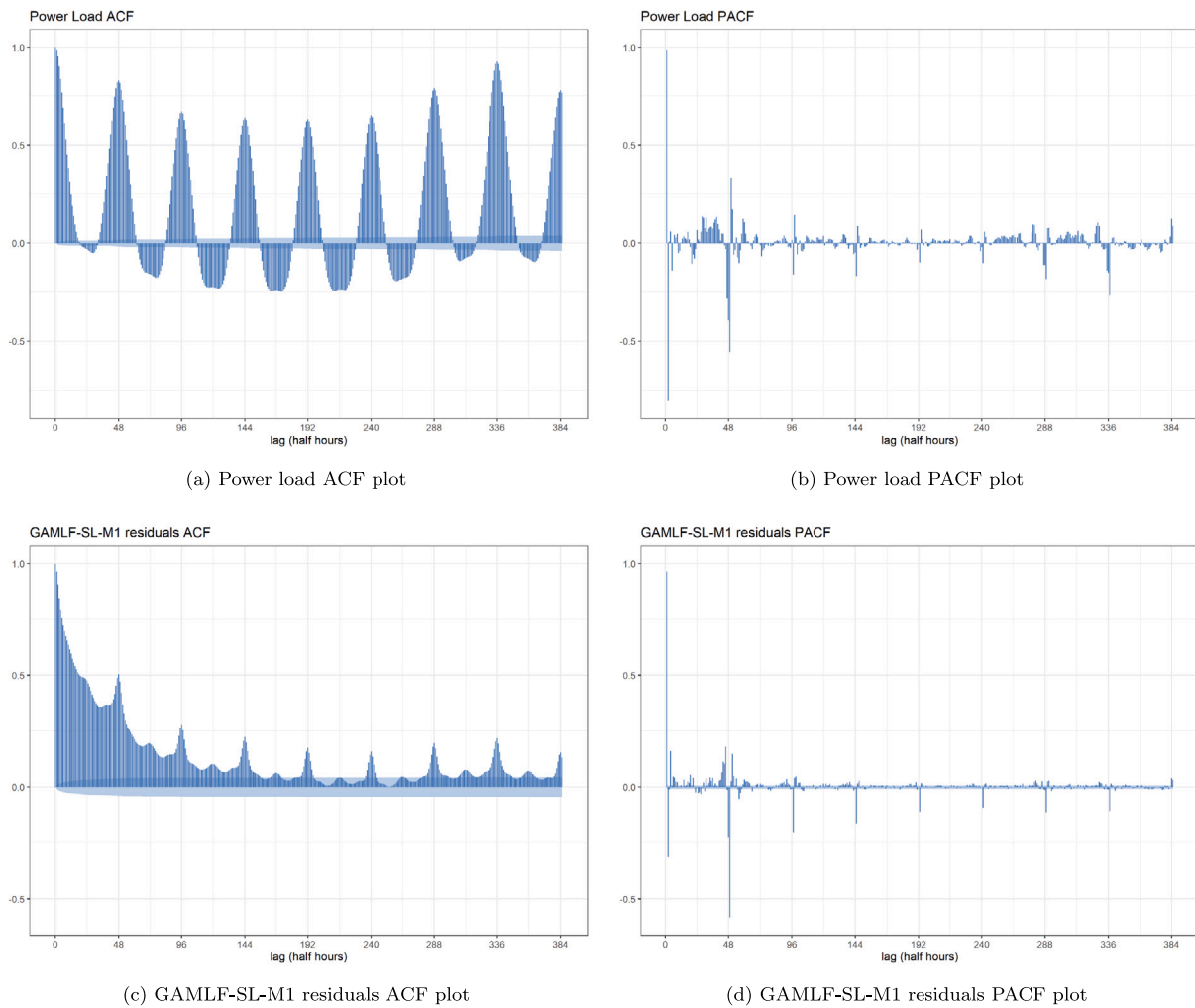


Fig. 5. ACF and PACF plots show the correlation of power load 5(a) and 5(b) or GAMLF-SL-M1 residuals 5(c) and 5(d) with a delayed copy of itself as a function of delay (8 days maximum). In both plots, high coefficients rise at immediately lag values and, more important, at lags of 48 (24 h), 96 (2 days), 144 (3 days), and so on. The amplitude of ACF of subsequently peaks decreases until 4 days lag (192 half hours) but increases again until reaching a new maximum amplitude precisely at 7 days lag (lag 336). Notice that the coefficients (columns) lie beyond the light blue region as they signify strong statistical confidence.

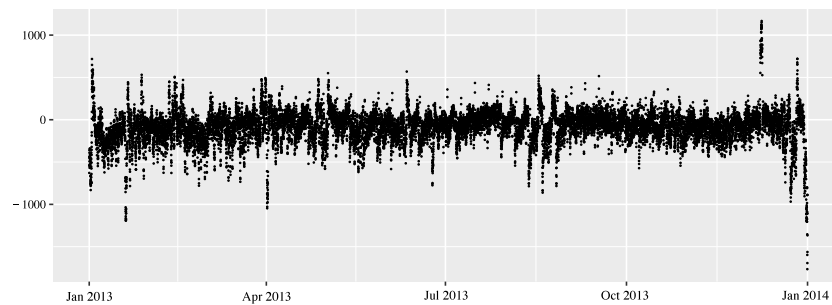


Fig. 6. Residuals over a year are analyzed to check patterns and obtain insights. Note the residuals in summer holidays – August –, Christmas and New Year holidays, and surrounding other public holidays.

of classes of 7 days of week to 9 day types, adding the class *Holiday-OnWeekend* for public holidays that occur on the weekend and *Holiday* otherwise.

Finally, after evaluating, the final model, GAMLF-SL-M3, is defined as

$$\begin{aligned} \hat{y}_t^{(M3)} = & \beta_0 + \beta_1 x_t^{(\text{Trend})} \\ & + f(\text{LagLoad24h}) (y_{t-48}) + f(\text{LagLoad1w}) (y_{t-336}) \\ & + f(\text{TimeOfDay}) (x_t^{(\text{TimeOfDay})}) \\ & + \sum_{i \in G} \mathbf{1}_{(x_t^{\text{DayType}} = i)} \left(\beta_i + f_i^{(\text{TimeOfDay/DayType})} (x_t^{(\text{TimeOfDay})}) \right) \\ & + \sum_{j \in \{\text{NewYear}, \dots, \text{Christmas}\}} \mathbf{1}_{(x_t^{\text{PublicHoliday}} = j)} \beta_j \\ & + f(\text{DayOfYear}) (x_t^{(\text{DayOfYear})}) \\ & + f(\text{Temperature}) (x_t^{(\text{Temperature})}) \\ & + f(\text{Temperature/TimeOfDay}) (x_t^{(\text{Temperature})}, x_t^{(\text{TimeOfDay})}) \\ & + f(\text{Temperature/DayOfYear}) (x_t^{(\text{Temperature})}, x_t^{(\text{DayOfYear})}) \\ & + \epsilon_t \end{aligned} \quad (5)$$

where $G = \{\text{MondayNoHoliday}, \dots, \text{SundayNoHoliday}, \text{Holiday}, \text{HolidayOnWeekend}\}$.

Note that $f_i^{(\text{TimeOfDay/DayType})}$, $f(\text{Temperature/TimeOfDay})$, and $f(\text{Temperature/DayOfYear})$ introduce the co-interactions appropriately, excluding their main effects. For instance, $\forall_{i \in G} f_i^{(\text{TimeOfDay/DayType})}$ smooth functions consider the intra-day effects over the power load differently for each i -day-type, but those have already excluded the main intra-day effect regardless of the day of the week. This latter effect has already been captured in the $f(\text{TimeOfDay})$ component. The same happens to $f(\text{Temperature/TimeOfDay})$ and $f(\text{Temperature/DayOfYear})$ components: those components capture the intra-day and annual seasonality effect of temperature considering that the main effect of temperature is already set up in the $f(\text{Temperature})$ component.⁵

To fit the models, the package *mgcv* R was adopted using thin plate regression spline bases, once they can smooth any number of covariates, avoid defining knots, and have some optimal properties (view sub-chapter 4.1.5 and table 5.1 from [67] for more information). Although effective degrees of freedom are controlled by the degree of penalization selected during fitting, by Generalized Cross-Validation, the upper limit⁶ on degrees of freedom is explicitly defined within each additive component [68].

2.4. GBMLF-SL

In the space of ensemble methods, gradient booster machines (GBM) have been a prominent technique used in a diverse number of machine learning and data mining challenges with considerable success [69]. The main idea of boosting is to add new learners to the ensemble formation sequentially. At each step, a new weak base-learner model is trained with respect to the error of the whole ensemble. Thus, the new base-learner formation is correlated with the negative gradient of the loss function associated with the whole ensemble [70,71].

The flexibility makes GBM highly customizable to diverse machine learning tasks. It has been considered to energy domain data-driven

tasks, including electricity load forecasting [72–74]. Hence, GBM's success in a disparate domain encourages its assessment and comparison with the models of previous sections.

The gradient boosting machine load forecasting – GBMLF-SL – is also a regression model for system level, as previous models already evaluated. The assessment is conducted using the same performance metrics and the same features which model GAMLF-SL-M3c used, though an adequate pre-processing is needed to apply GMB. Three different base learners and explored: (i) tree-based model, (ii) linear functions with L_1 and L_2 regularization, and (iii) component-wise smoothing splines.

To the vector feature $\mathbf{x}_t$ is applied adequate pre-processing, as temperature data are explicitly unfold to three exponential versions (the temperature raised to the power of 1, 2 and 3), the 1-day and 1-week lagged loaded are also explicitly included, categorical features are encoded into binary variables using one-hot encoded, and the numerical variables are individually scaled and centered (once the linear base learner with L_1 and L_2 regularization are sensible to different scales between features).

The Extreme Gradient Boosting algorithm (XGboost) was used with a hyperparameter exhaustive grid search.

2.5. GAMLF-SLE

The GAM based load forecasting, GAMLF-SLE, is a regression model for system level resulting from the ensemble of a set learners based on the previous section. Ensemble methods can be applied to national load forecasting to improve accuracy. There are periods like the August summer break and days around public holidays that are particularly challenging in forecasting tasks.

Ensemble methods get a collection of learners and combine their outcomes to make an overall decision. The rationale behind this is due to the fact that different learners, often referred to as experts, might be more accurate on specific sub-problem, period time, or category. Therefore, a mechanism to build different learners that can capture specific information or patterns in the data must be defined.

The Weighted Majority Algorithm (WM) is a simple and effective method, based on weighted voting, to build up a compound algorithm from the outputs of some pool of known algorithms where at least one of them will perform well [75]. Algorithm 1 is the variant of the original WM algorithm that was changed to meet the following requirements:

- be able to support numerical and unbounded outcomes from predictors, using a linear mapping to predictors' outcomes before entering the Weighted Majority Continuous (WMC) Algorithm, and whose output, in turn, is mapped back to original domain;
- be able to receive the label with a constant time delay, δ , and hold back the learning process, in this case 48 trials later (one-day);
- be able to learn in independent and predefined spans, called regime r , where each one is a discontinuous and disjoint temporal period, and thus some predictors will be designed and calibrated exclusively for specific regimes;
- be able to cope with non-available outcomes of a subset of predictors and update the weights in accordance.

2.6. GAMLF-SSL

The GAM based load forecasting, GAMLF-SSL, is a regression model for secondary substations level. The approach aims to predict the power load on thousands of assets at a level more disaggregated when compared to previous section. The more disaggregated the forecasting, the more complex and difficult it is to predict. There is literature related to how to build forecasting models to a national or regional electricity consumption, but much less literature is available for medium-

⁵ For reproducibility in R's *mgcv* package consider the formula $y \sim \text{trend} + s(\text{timeOfDay}, k = 40) + \text{ti}(\text{timeOfDay}, \text{by} = \text{dayOfWeek}, k = 20) + \text{dayOfWeek} + s(\text{dayOfYear}, k = 40) + s(\text{temperature}, k = 30) + \text{ti}(\text{temperature}, \text{timeOfDay}, k = 7) + \text{ti}(\text{temperature}, \text{dayOfYear}, k = 7)$.

⁶ Argument k which appears in the formula for the package *mgcv*.

Algorithm 1: Weighted Majority Algorithm — Continuous Reviewed Version.

Let N be the number of forecasters, and R the number of regimes.
 Choose the learning parameters $\eta_r > 0$ that would be used in the weight update rule.
 Choose the time delay parameter $\delta > 0$.

```

1: Initiate weights  $\omega_{i,r}^{(1)} \leftarrow 1 \quad \forall i \in 1 \dots N \quad \forall r \in 1 \dots R$ 
2: for  $t \leftarrow 1 \dots T$  do
3:   Receive instance  $x \in \mathcal{X}$  of regime  $r \in 1 \dots R$ 
4:   Receive forecasts  $y_1 \dots y_N \in [0, 1]^N$ 
5:   Predict  $\hat{y} \leftarrow \frac{\sum_{i \in 1 \dots N \wedge y_i \text{ exists}} \omega_{i,r} y_i}{\sum_{i \in 1 \dots N \wedge y_i \text{ exists}} \omega_{i,r}} \in [0, 1]$ 
6:   if  $t > \delta$  then
7:     Let  $r'$  be the regime at time  $t - \delta$  that is  $r' \equiv r^{(t-\delta)}$ 
8:     Let  $\hat{y}'$  be the prediction at time  $t - \delta$  that is  $\hat{y}' \equiv \hat{y}^{(t-\delta)}$ 
9:     Receive true label  $y' \equiv y^{(t-\delta)} \in [0, 1]$ 
10:    Incur loss  $\ell_{\text{abs}} \leftarrow |\hat{y}' - y'|$ 
11:    Update weights  $\omega_{i,r'}^{(t+1)} \leftarrow \omega_{i,r'}^{(t)} \exp^{-\eta_{r'} \cdot |\hat{y}'_i - y'|} \quad \forall i = \{i \in 1 \dots N : \hat{y}'_i \text{ exists}\}$ 
12:    Update weights  $\omega_{i,r'}^{(t+1)} \leftarrow \omega_{i,r'}^{(t)} \exp^{-\eta_{r'} \cdot \ell_{\text{abs}}} \quad \forall i = \{i \in 1 \dots N : \hat{y}'_i \text{ not exists}\}$ 

```

and low-voltage forecasting (except for individual low-voltage clients). However, the same principles are applicable to the disaggregated STLF with the appropriate changes and further extensions.

The structure of the model is based on Eq. (5). The forecast of the power load at time t for the asset n is⁷

$$\hat{y}_{t,n} = \begin{cases} \beta & |S_n| = 1 \\ \hat{y}_{t,n}^* & |S_n| > 10 \\ \hat{y}_{t,n}^* - \underbrace{f(\text{LagLoad24h})(y_{t-48,n}) - f(\text{LagLoad1w})(y_{t-336,n})}_{\text{covariates removed}} & \text{otherwise} \end{cases} \quad (6)$$

where $\hat{y}_{t,n}^*$ is the same Eq. (5) without the trend covariate,

$$\hat{y}_{t,n}^* \stackrel{\text{def}}{=} \hat{y}_t^{(\text{M3})} - \beta_1 x_t^{(\text{Trend})}. \quad (7)$$

Due to the capabilities of the energy meters, the measurements are actually non-negative integers, $y_n \in \mathbb{N}_0^T$. However, this fact was not considered during the regressor modeling and during its accuracy assessment. There are assets that present a constant measurement $y_n \in \{c\}^T$, or a small set of unique values, a small $|S_n| = |\{y_{t,n}\}_{t \in \{1, \dots, T\}}|$. For those timeseries, a simpler regressor structure was used since the number of free parameters to estimate cannot be larger than the number of unique observations. Therefore, the third element of Eq. (6) removes the covariates related to lagged load.

3. Experiments and analysis

3.1. Evaluation criteria

Besides the accuracy criteria on which most articles focus, the models are evaluated from three more aspects. These are the four aspects:

- **Applicability**—The model uses data and information that the utility is able to obtain with tangible resources (both human, data, and system resources). For example, if a utility is unable to get an up-to-date calendar of special events, such as strikes, conferences, sports, and cultural events, which have an impact on the power load at the secondary substation that feeds the infrastructure or venue, then a model containing that variable is not applicable, no matter how well that variable would improve the predictions.

⁷ Actually, for implementation purposes, this model is used when the original model $\hat{y}_{t,n}^*$ returns an error due a small (undefined) $|S_n|$.

- **Interpretability**—The degree to which a human can understand (mental model) and consistently deduces the result of the model [76]. When the model is more interpretable, humans can more easily understand why certain outcomes are realized and consequently accept them easily. Note that the term *explainability* is used rather to refer to the explanations of individual predictions.
- **Reproducibility**—The model and the method carried out to build the model are systematic and documented, and it is possible to replicate them in other time horizons, geographies, or grid levels. If engineering heuristics are involved, the tweak and tuning process should be well defined.
- **Accuracy**—The performance of the model in terms of how close the predictions are to the real values. In particular, the mean absolute error (MAE), the mean absolute percentage error (MAPE), the root mean square error (RMSE), the normalized RMSE (NRMSE), the coefficient of determination R^2 , and the mean absolute scaled error (MASE) [77]. Note that these accuracy measures focus on smoothness of the forecasts, rewarding them. At the distribution level, it is the peak that matters for many use cases. So, for secondary substation models, we used Haben's adjusted error and a normalized version that prefer models that predict peaks even within a restricted displacement that do not forecast at all [78]. Additionally, the timeseries cross-validation procedure is used to assess performance, that is, the corresponding training set consists only of observations that occurred before the observation that forms the test set [79].

Let y_t be the real value and $\hat{y}_t$ the prediction at time t , define the error metrics for the period $[1, T]$ as

$$e_{\text{MAE}} = \frac{1}{T} \sum_t |y_t - \hat{y}_t|, \quad (8)$$

$$e_{\text{MAPE}} = \frac{1}{T} \sum_t \left| \frac{y_t - \hat{y}_t}{y_t} \right|, \quad (9)$$

$$e_{\text{RMSE}} = \sqrt{\frac{1}{T} \sum_t (y_t - \hat{y}_t)^2}, \quad (10)$$

$$e_{\text{NRMSE}} = \frac{\sqrt{\frac{1}{T} \sum_t (y_t - \hat{y}_t)^2}}{\frac{1}{T} \sum_t y_t}, \quad (11)$$

$$e_{R^2} = \frac{\sum_t (\hat{y}_t - \bar{y})^2}{\sum_t (y_t - \bar{y})^2}, \quad (12)$$

$$e_{\text{MASE}} = \frac{\sum_t |y_t - \hat{y}_t|}{\sum_t |y_t - y_{t-m}|}, \quad (13)$$

Table 6

GLMLF-B results calculated over the time span from 2016 and 2019 using timeseries cross-validation with a fixed 3-year window for training and the next 1 year, 2 weeks, 1 week or 1 day of testing data. The best results are in bold. The MASE is calculated with m steps equal to 52 weeks, 2 weeks, 1 week, and 1 day, respectively. All metrics are the mean calculated from cross-validation folds.

Update Cycle	Folds	MAE (MW)	MAPE (%)	RMSE (MW)	NRMSE (%)	R ²	MASE
1 year	1	235.25	4.23	353.12	6.14	0.888	0.519
2 weeks	26	217.75	3.87	296.99	5.17	0.918	0.742
1 week	52	214.64	3.82	283.73	4.96	0.929	0.970
1 day	365	210.46	3.75	249.74	4.44	0.958	1.207

where m is the seasonal period for executing the naive m -step seasonal forecast method, $\hat{y}_t^{\text{NAIVE}} = y_{t-m}$. If the timeseries presents more than one seasonality, it is compared to the seasonality with the lowest period. This latter error metric is scaled-free and is well suited, unlike the MAPE, to timeseries with zero or near zero values because it never gives infinite or undefined values, except in the irrelevant case where $\forall_t y_t = C$.

Haben's adjusted p -norm error (APN), the mean adjusted p -norm error (MAPN), and the normalized mean adjusted p -norm error (NMAPN), are defined as

$$e_{\text{APN}} = \min_{\mathbf{P} \in \mathcal{P}(w)} \|\mathbf{P}\hat{\mathbf{y}} - \mathbf{y}\|_p, \quad (14)$$

$$e_{\text{MAPN}} = \sqrt[p]{\frac{1}{T} e_{\text{APN}}^p}, \quad (15)$$

$$e_{\text{NMAPN}} = \frac{\sqrt[p]{\frac{1}{T} e_{\text{APN}}^p}}{\frac{1}{T} \sum_t y_t}, \quad (16)$$

where $\mathcal{P}(w)$ is the complete set of restricted permutations such that $P_{ij} = 0$ for $|i - j| > w$, restricting the magnitude of the displacements of the forecast values. Haben suggests using the absolute 4-norm error, $p = 4$, to penalize large errors (i.e., missed peaks) much more than small errors. The choice of the adjustment limit w depends on the use case. Nevertheless, for general purposes, this study uses $w = 3$, which means that forecasts can be displaced by up to 3 half hours on either side of their original forecast time.

3.2. GLMLF-B results

The generalized linear model-based load forecasting—benchmark model, GLMLF-B, is evaluated using dataset I, whose results are shown in Table 6 for different update cycles. The metrics are computed through the following procedure: (i) over the time span from 2016 and 2019, take the three consecutive years as training data and the next period of updating cycle (one day, one week, two weeks, and one year) as testing data; (ii) roll the actual data of this testing period to the training data and recalculate the model; (iii) recompute the metrics with this new model, and so on, until all the periods in the year 2019 are forecasted. Table A.10 has the detailed metrics including minimum, maximum, second, and third quantiles, as well as median and mean for timeseries cross-validation folds.

The GLMLF-B model uses trend index, calendar, time, and temperature variables. All variables are acceptable by domain experts because of their interpretability: a quantitative variable (trend) to capture the increasing or decreasing trend of the power load over the interval, calendar variables to capture different effects in months and days of the week, time to capture intraday seasonality, and the effect of temperature on the power load. The model is highly interpretable because the weights, which result from the fitting process, manifest the degree of importance or effect of each variable. For example, one of the weights shows the effect of i th hour on the power load and how that compares to the same hour on the previous day of the week.

From the applicability perspective, the use of a trend index obligates the refitting of the model as periodically as the change of that trend. Otherwise, the performance of the model rapidly degrades. Moreover,

the use of temperature values observed at the exact time step for which the power load is forecast forces the availability of the temperature forecast as input to the model. Thus, temperature prediction errors could potentially decrease the performance of the power load predictions traced in this study.

3.3. GAMLF-SL results

The generalized additive model-based load forecasting—system-level models, GAMLF-SL, are evaluated using the same dataset I and procedures described in former section. The results for GAMLF-SL are shown in Tables 7 and A.10 for the same update cycles. As a first result, it is obvious that GAMLF-SL-M1 achieves a better accuracy compared to the results of the benchmark model, GLMLF-B. Taking into account the same or similar variables, the GAM technique achieves better accuracy with simplified interpretability (discussed later). Moreover, the systematic approach iteratively enhances the GAM-based regression models by introducing new synthetic explanatory variables based on domain knowledge. GAMLF-SL-M2 adds autoregressive components to the previous model, and GAMLF-SL-M3 in turn adjusts and adds calendar features. When comparing the benchmark model, GLMLF-B, and the enhanced GAMLF-SL-M3 model, MAPE, MAE, RMSE and NRMSE were reduced by 42% to 47% in all update cycle scenarios. Furthermore, as the median and average accuracy improves, the remaining days with higher errors also improve in terms of error amplitude. This is demonstrated by the decrease in the higher errors exhibited in the MAPE and MAE box plots (Fig. 7) for the four models cross-validated with 365 folds, each representing one day of testing. Fig. 8 shows the improvement on special days as summer warmer days in June, July and August, as well as, weekends and holidays.

With respect to interpretability, the internals of the model are easy to interpret. Besides the parameters β , the spline functions f , which are also estimated during the fitting step, are easy to understand and familiar. The fact that experts see their domain knowledge “recognized” by the model reinforces acceptance and adoption. Fig. 9 shows the main effect of the intra-day pattern 9(a), as well as the interaction of the same effect for each day of the week after the main effect being appropriately excluded 9(b)–9(i). Note that the model adds a lower load baseline for weekends compared to working days (Fig. 9(b)). Consequently, to maintain the same levels of power load during the very early morning (0 a.m. to 6 a.m.), as on any other ordinary day, f_{Saturday} 9(h) and f_{Sunday} 9(i) display, for that period, higher responses compared to other spline functions 9(c)–9(g). Thus, this compensates for the low load baseline added by β_{Saturday} and β_{Sunday} 9(b). On the other hand, the peaks and valleys of those two weekend spline functions are different from those of the working days. Even the number of effective degrees of freedom is higher for those two days due to the more complex structure required to reshape the baseline $f^{(\text{TimeOfDay})}$. Note that the baseline multiplied by β_i , which is always added, gives a more approximate shape to the load curve of the working day.

Furthermore, β_1 , which results from the linear component of the timeseries, and $f^{(\text{DayOfYear})}$, which emerges from the annual pattern, are shown in Fig. 10. Note the pronounced valley shape at the beginning and end of the year, as well as in the summer leave (August): they suggest a trend as lower electricity consumption. The effect of temperature

Table 7

GAMLF-SL results calculated over the time span from 2016 to 2019 using timeseries cross-validation with a fixed 3-year window for training and the next 1 year (1-fold), 2 weeks (26-fold), 1 week (52-fold) or 1 day (365-fold) of testing data. The best results are in bold. MASE was calculated with m -steps equal to 52 weeks, 2 weeks, 1 week, and 1 day, respectively. All metrics are the mean calculated from cross-validation folds.

	Model	MAE (MW)	MAPE (%)	RMSE (MW)	NRMSE (%)	R ²	MASE
1 year	M1	227	4.09	323	5.61	0.909	0.501
	M2	141	2.53	229	3.99	0.948	0.311
	M3	127	2.26	191	3.32	0.965	0.280
2 weeks	M1	202	3.59	276	4.81	0.927	0.699
	M2	138	2.47	198	3.48	0.944	0.468
	M3	124	2.20	172	3.01	0.963	0.433
1 week	M1	200	3.56	265	4.64	0.933	0.907
	M2	138	2.46	190	3.35	0.946	0.603
	M3	124	2.19	166	2.91	0.964	0.568
1 day	M1	194	3.46	234	4.16	0.957	1.142
	M2	137	2.44	166	2.97	0.967	0.686
	M3	123	2.18	148	2.64	0.977	0.653

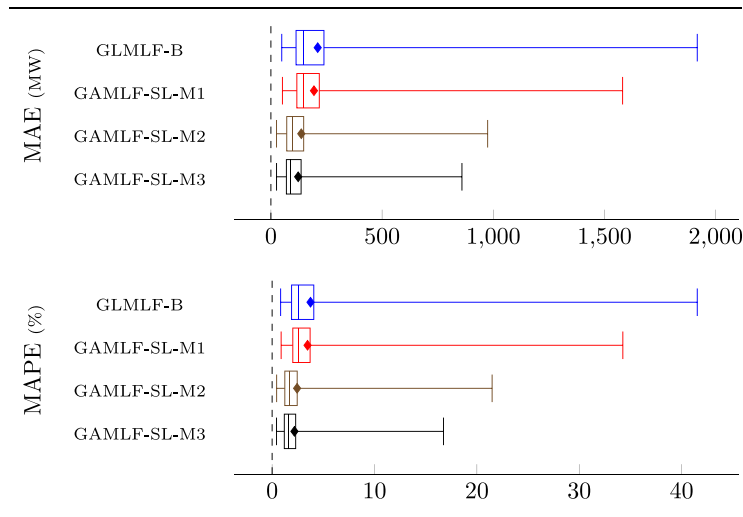


Fig. 7. Quantile box plots of MAPE and MAE for the four models cross-validated with 365 folds (1 day update cycle). Not only do the median and average accuracy improve, but days with higher errors are also predicted more accurately.

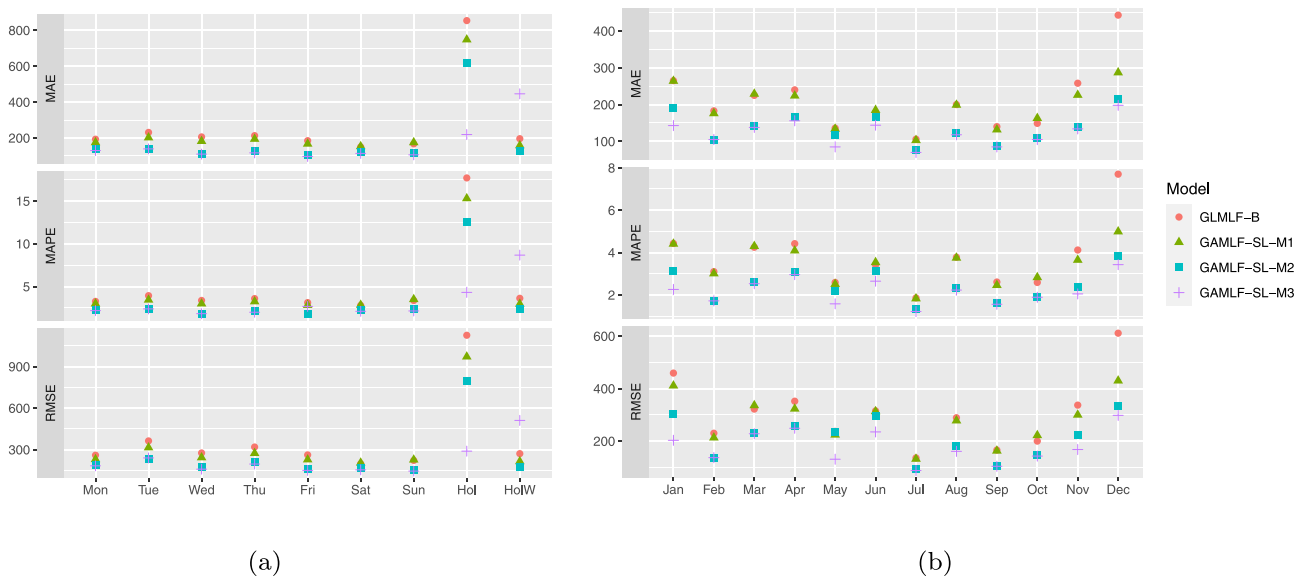


Fig. 8. MAE, MAPE, and RMSE by month or day type calculated for the four models cross-validated with 365 folds (1 day update cycle). For the day types “Hol” means holiday and “HolW” means holiday on a weekend. The day types are disjointed subgroups.

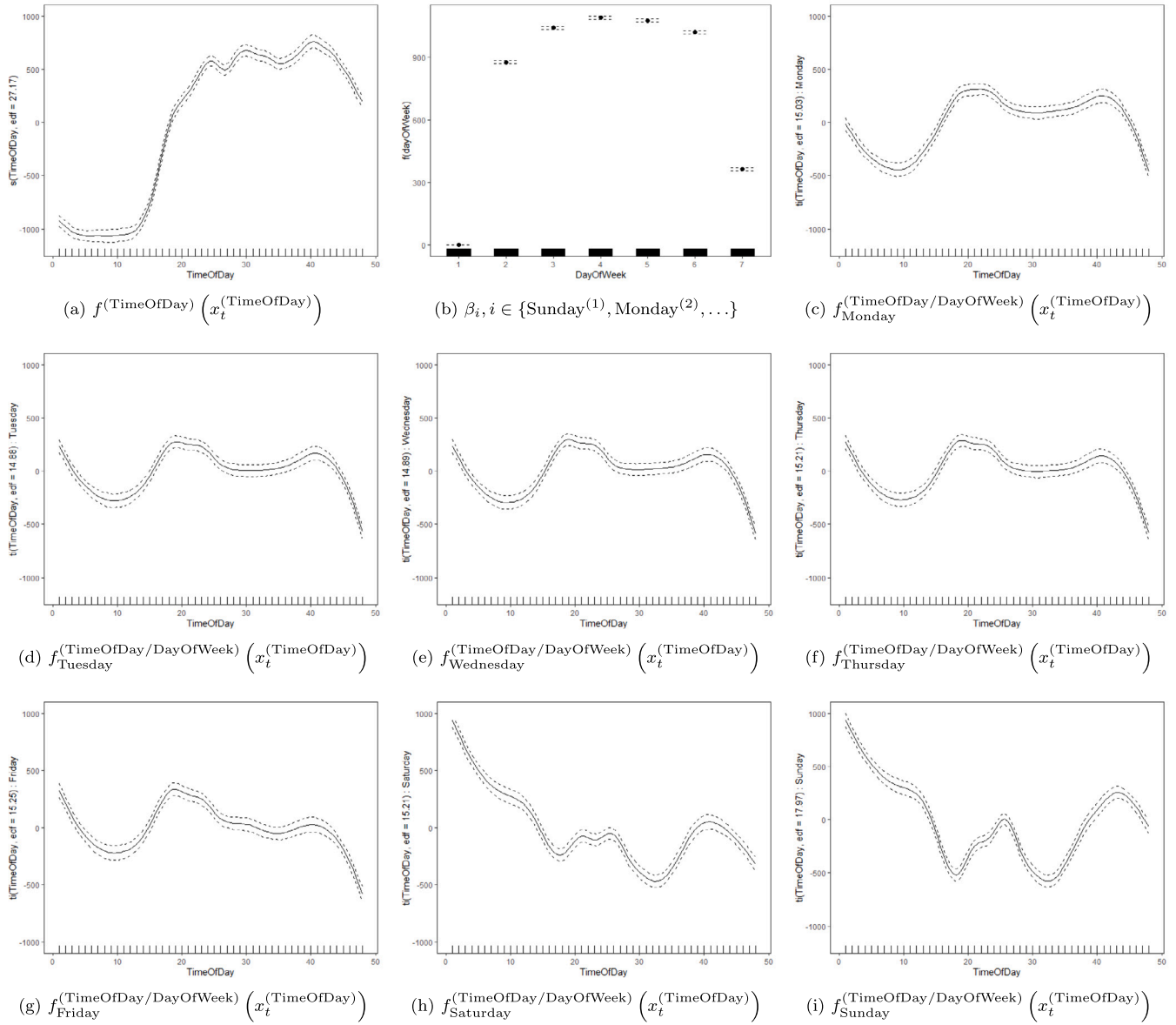


Fig. 9. The effects of intra-daily and weekly patterns on the power load are recognized by the parameters β and spline functions f which were estimated during the fitting step of the GAMLF-SL-M1 model. The figure (a) exhibits the main effect of intra-day pattern independently of the day of week, while figures (c–i) exhibit the effect of intra-day pattern for each day of the week after the main effect being appropriately excluded.

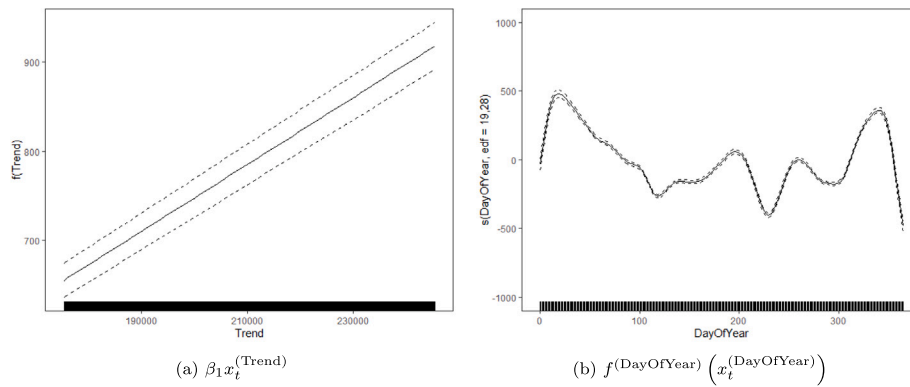


Fig. 10. The linear component and the annual pattern recognized by the β_1 parameter and the $f^{(\text{DayOfYear})}$ spline function, which are estimated during the fitting step of the GAMLF-SL-M1 model.

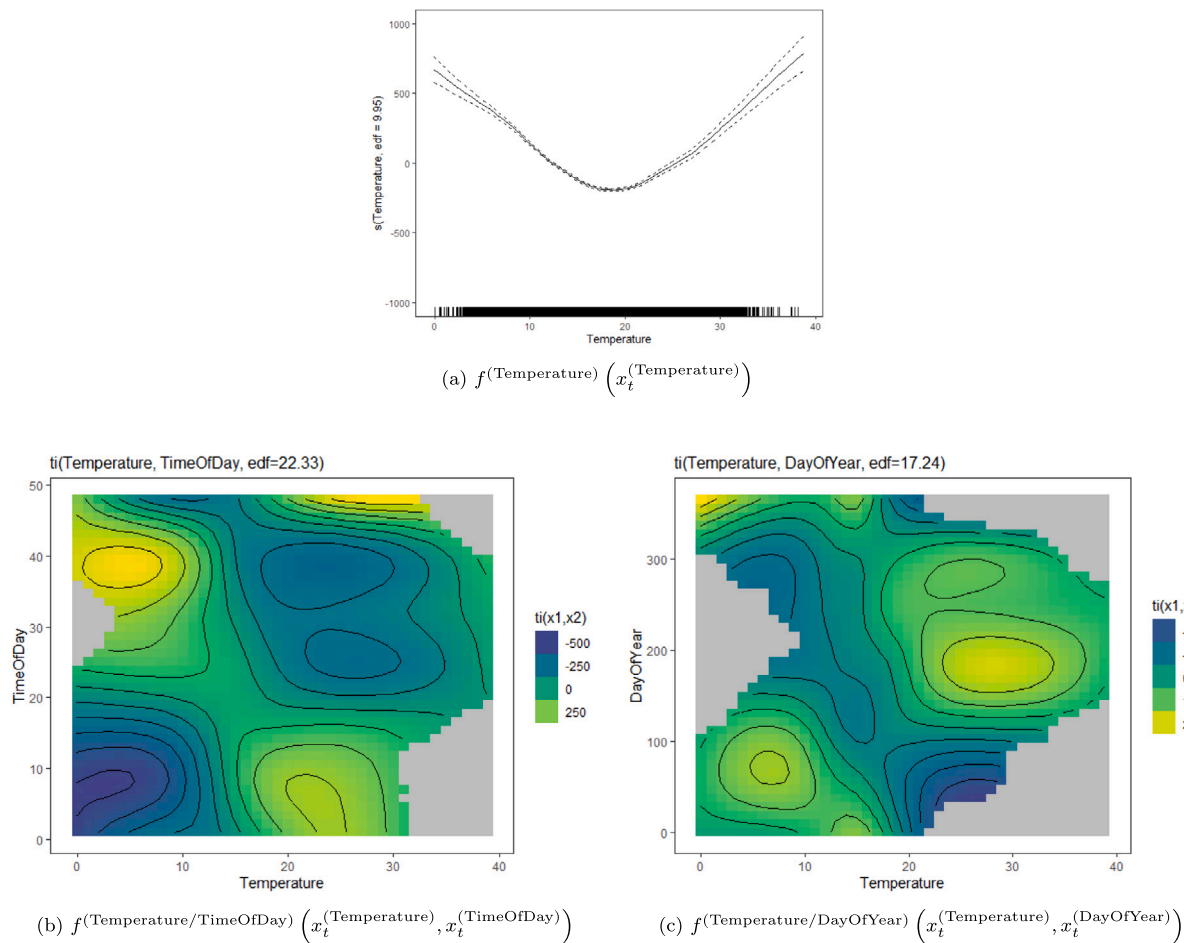


Fig. 11. The effects of temperature over the power load recognized by the spline functions f which were estimated during the fitting step of model GAMLf-SL-M1. Figure (b) and (c) represent the bivariate effect of the temperature and the time of the day (b) or the day of the year (c).

on the power load is highlighted in the graphs in Fig. 11. The main effect of temperature 11(a) is aligned with the scattered plot in Fig. 3. It can be seen from 11(b) that low temperatures during the early morning (0 am to 6 am) have the opposite effect to the same low temperature during the evening. During the evening, low temperatures reinforce the general pattern 3, that is, increase the power load. On the contrary, during the early morning, low temperatures shrink the effect of that same general pattern. Another insight is that warm winters have the effect of shrinking load predictions due to a reduction in heat demand 11(c).

More insights could be derived from the internals of the fitted model, the point is: the model proposes a highly interpretable internal structure.

3.4. GBMLF-SL results

In this section, the gradient booster machine-based load forecasting—system-level model, GBMLF-SL, is evaluated. The same dataset I and the procedures to compute the results are followed as in the previous section.

The results for GBMLF-SL indicate no improvements in accuracy when comparing the RMSE with the previous model, GAMLf-SL. The 199MW RMSE achieved compares to the 191MW RMSE of GAMLf-SL-M3, the latter better.

Given the same data modeling, even with some tweaks to accommodate the characteristics of the gradient booster machine (GBM), it achieves the same accuracy, but with two disadvantages to be noted.

GBM has many hyperparameters and they need to be optimized by an outside optimization algorithm that figures the hyperparameters based on minimization of loss. That takes computing time and effort to choose the optimization algorithm and define the grid search, though it is easily parallelizable.

Furthermore, GBM ensemble structure are notably less interpretable compared to GAM, and the purposed evaluation of the models ensures the qualitative balance between their accuracy and interpretability. Although weak base learners are known as highly interpretable as linear functions or small-sized trees, to achieve the same levels of accuracy, GBM holds thousands of linear functions or trees weightily combined to formulate the ensemble, losing the interpretability from the domain perspective. Nevertheless, model-agnostic interpretation techniques for machine learning models such as partial dependence plots (PDP), permutation feature importance (PFI) and Shapley values provide insightful model interpretations [80–82] if well used [83].

3.5. GAMLf-SLE results

Several learners, also known as experts, are combined into a master model using the WMC variant described in Algorithm 1. It must be parameterized by the delay constant δ and the learning constant η_r , which is optionally dependent on the actual regime. Once the goal is to forecast 24 h ahead, the real label is collected 24 h after the prediction occurs. Therefore, the delay constant δ has a value of 48 half-hours. The learning constant η_r is set as one for every regime r . Additionally, regimes are defined annually as discontinuous and disjoint temporal

Table 8

Years are partitioned into several discontinuous and disjoint intervals called regimes. Each forecasting model might have better accuracy in a specific regime.

Regime	Description
R1	Christmas and New Year period (23 December to 2 January)
R2	Carnival period (Sunday to Wednesday including Tuesday Carnival)
R3	Easter period (Thursday to Monday including Easter Sunday)
R4	Other public holidays
R5	Weekends (except previous regimes)
R6	August (Summer holidays except weekends and public holidays)
R7	Other common days during Spring and Summer (except August)
R8	Other common days during Autumn and Winter

Table 9

Both general-purpose and period-specific models were combined into a master model to minimize error (RMSE and MAPE). Units of RMSE are MW.

Model	RMSE (MW)	MAPE(%)	Model introduced
A	203.26	2.580	General-purpose model (Eq. (7))
B	179.54	2.313	General-purpose model reviewed (including covariate that takes into account the demand 1 week ago)
C	169.74	2.182	Weekends' model
D	163.91	2.094	August's model
E	165.03	2.083	Public holidays' model
F	162.06	2.046	Spring and Summer's model
G	159.53	2.025	Easter's model
H	159.90	2.027	Carnival's model
I	154.04	2.004	Christmas and New Year's model
J	158.88	2.073	Autumn and Winter's model

intervals. When experts are combined, the weights learned are independent of the regimes. Table 8 summarizes the regimes used throughout the year. The algorithm starts learning (update the weights) from the (6 years long) training dataset and continues over the (2 years long) testing dataset. The accuracy of the algorithm is only based on the results of the last two years.

The experts are fitted with minor changes either by modifying the features of regression model or by fitting it against picked temporal periods, which comprise different training data. Eq. (7) is used as the first expert model A, and model B is achieved by adding the 1 week lagged load. Finally, different training datasets are used to fit the same model structure, resulting in different experts: models C–J. The training dataset is chosen according to the defined regimes and its temporal intervals. As shown in Table 9, it resulted in a set of predictors that perform well under specific periods or conditions. To evaluate the master model, the global RMSE and the global MAPE are also calculated in each iteration cycle after combining the newest predictor with the master model. The ensemble model I achieves better performance than the initial model A. The ensemble model's accuracy for one-day ahead forecast is about 154 MW for the global RMSE and 2.0% for the global MAPE.

The modified weighted majority algorithm (continuous version) proved to be an effective method to achieve better accuracy compared to a general-purpose predictor for STLTF when GAM was used as a base technique. Although the predictors, which were combined, use the same regression technique – which is not required –, they have different specificity goals. Some predictors were specifically calibrated for special days or yearly/weekly repeatable periods and, therefore, they are more accurate on those specific days compared to general-purpose ones. Note that not only the regression technique was the same for all predictors, but also no new explanatory variables were added. Therefore, we conclude that the combination of the WMA while ensemble technique and the GAM while base modeling technique for predictors improves the overall accuracy. However, the WMC learning process includes no regulation, and thus overfitted predictors can adulterate the weights of the ensembling.

From the applicability perspective, though this method uses more data to train predictors, plus the initial ensemble training process, this is online learning. So, the method would decrease the importance of a specific predictor if its accuracy decays over time and vice versa. It is also important to remark that the ensemble layer provides an

interpretable configuration: the weights impose the importance of each model within a regime, and the regimes are highly understandable by knowledge domain and acceptable by operators and managers. Therefore, this ensemble method does not decrease the degree of interpretability obtained from the basis predictors.

3.6. GAMLF-SSL results

To assess the performance of $\hat{y}_{t,m}$, the generalized linear model-based load forecasting at the secondary substation level, the entire year 2019 was used as a test (see Fig. 12 for an example of an asset). Note that one year is the minimum acceptable to test a forecasting model whose target value shows annual seasonality. There are several assets whose data was not available before 2019 for training and testing purposes. So, the following results consider the forecasting models for 22,974 PTC assets and 61,689 PTD assets.

Another relevant issue is the prediction horizon, which was set at 24 h (48 half-hours). This means that for any time $t + 48$ the model forecasts, the available explanatory variables refer to a time at or before t , except for the temperature predictions.

To accurately assess the performance of the forecasting models, several error metrics (MAE, MAPE, RMSE, NRMSE, R^2 , MASE, APN, MAPN, NMAPN) were used for each asset forecast $\hat{y}_t$. Only the scaled errors, MAPE, MASE, NRMSE, and NMAPN, were passably analyzed together and presented as histograms.

Fig. 13 exhibits the MAPE histograms. Half of the models, which forecast the power load of the PTD, have a MAPE below 0.126, while only 16.35% of the PTC models present a MAPE below the same value. Both distributions are right-skewed, but the PTC error distribution tail is flatter than the PTD's. Therefore, the model structure is better designed to cope with the forecast of the PTD power load.

Fig. 14 shows the MASE histograms. Note that MASE compares the model with a naive forecast method. Taking into account daily seasonality, the naive forecast method returns the power load observed 24 h before, $\hat{y}_t^{\text{NAIVE}} = y_{t-48}$. The error metric is scale-free and suited to timeseries with zero or near zero values because it never gives infinite or undefined values except for the irrelevant case where $\forall_t y_t = C$. Both graphs also show a vertical line $e_{\text{MASE}} = 1$. Models are better than the naive model when $e_{\text{MASE}} < 1$. 82.8% PTD models and 66.0% PTC models are better than the naive model.



Fig. 12. The performance calendar plot shows which days have the highest residuals. Public holidays are spotted on the calendar with the respective symbols, as well as the time t that divides the training and testing datasets, marked by a horizontal line on the specific day (3 May 2017). Christmas and New Year, as well as the following day, show a higher error depending on the day of the week (weekend or not), which suggests that it is necessary to review the structure of the model in these combinatorial cases.

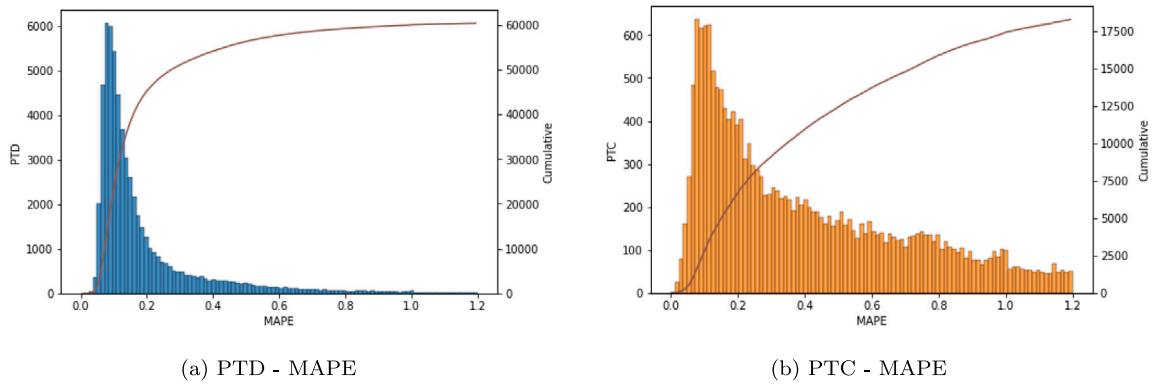


Fig. 13. Mean Absolute Percentage Error for PTD and PTC models. Both distributions are right-skewed but with different flatness. The brown line indicates the cumulative count. The graph was cut in $e_{MAPE} = 1.2$, which means that 1366 PTD models (2.21%) and 4692 PTC models (20.42%) are out.

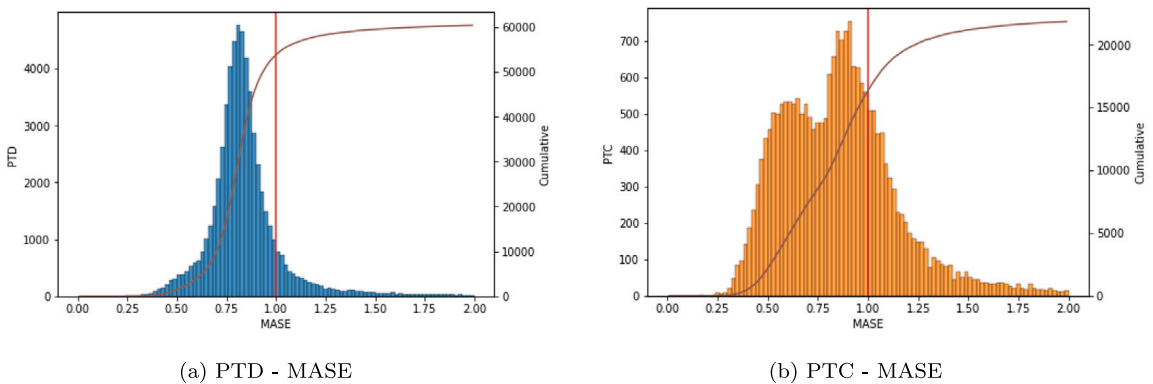


Fig. 14. Mean Absolute Scaled Error for PTD and PTD models compared with naive forecast, which returns the power load observed 24 h earlier. 53,391 PTD models (86.5%) and 16,083 PTC models (70.0%) are better than the naive model, $e_{MASE} < 1$. The brown line indicates the cumulative count. The graph was cut in $e_{MASE} = 2$, which means that 1308 PTD models (2.21%) and 1128 PTC models (4.91%) are out.

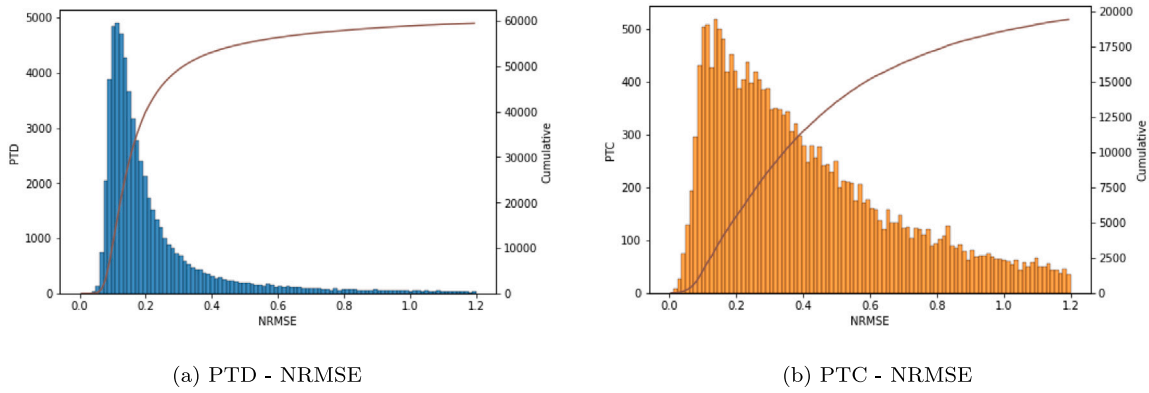


Fig. 15. Normalized Root Mean Square Error for the PTD and PTC models. Both distributions present a long tail. The brown line indicates the cumulative count. The graph was cut in $e_{\text{NRMSE}} = 1.2$, which means that 2214 PTD models (3.59%) and 3533 PTC models (15.38%) are out.

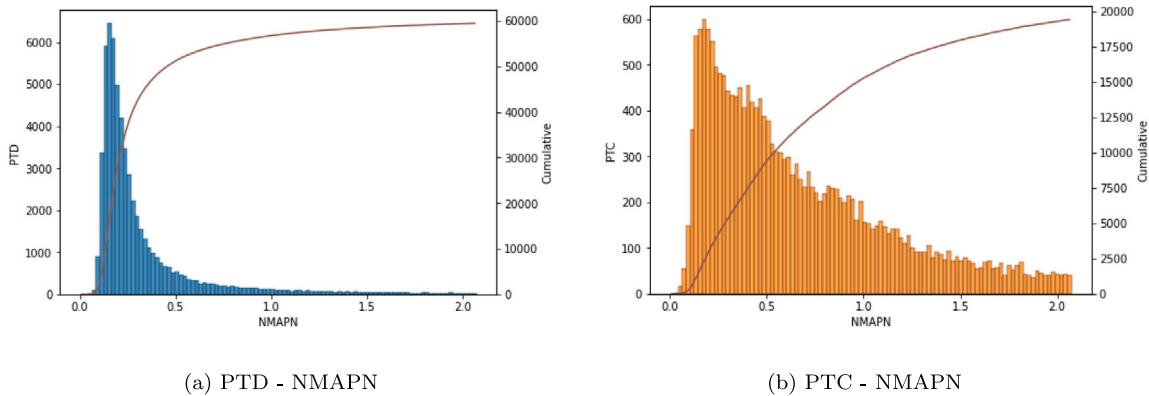


Fig. 16. Normalized Mean Adjusted p -norm Error with $p = 4$ and $w = 3$ for the PTD and PTC models. Both distributions present a long tail. The brown line indicates the cumulative count. The graph was cut in $e_{\text{NMAPN}} = 2.0736$, which means that 2257 PTD models (3.66%) and 3580 PTC models (15.58%) are out.

Fig. 15 shows the NRMSE histograms and Fig. 16 the NMAPN histograms. Half of the models that predict the PTD power load have a NRMSE below 0.162, while only 16.79% of the PTC models have an NRMSE below the same value. While, half of the models that forecast the PTD power load have a NMAPN below 0.222, while only 14.32% of the PTC models have an NMAPN below the same value. Both distributions show a long tail due to a percentage of assets that does not follow the general structure or the quality of timeseries is low (for instance, non-decimal values with a low amplitude).

In general, the model is capable of forecasting such diverse amounts of timeseries (active power load) for all 84,663 secondary substations of the Portuguese mainland grid, in which data was available to train and test. It was concluded that this model is better than the naive model in 82.1% assets. However, hypotheses have been put into place to be tested in further work.

3.7. Distributed system architecture

From an applicability perspective, the disaggregated forecast problem challenges the capacity of storage and computing to deal with hundreds of thousands of time series and artifacts, such as fitted models, feature data tables, images, metrics, and logs. The system architecture was designed considering important requirements for a feasible system that would be integrated into existing DSO business processes and promote the continuous innovation process in energy management. In fact, this resulted in an enterprise forecasting system called PREDIS—PREvisão DIStribuída whose results have been used to anticipate load peaks and network constraints [84].

A set of requirements was considered during the design and implementation of the system: (i) **big data storage** for load curves,

meteorological observations and/or predictions, and predictions from the forecast procedure, (ii) **computing scalability** to perform individual model fitting for each asset and its daily predictions in a useful time (a few hours to complete the forecast once the new data have been appended to time series), (iii) **responsibility segregation** design to cope with different perspectives (trustful data pipeline engineering, agile data science methodology, and parallel machine learning engineering), (iv) **traceability**—model versioned and lineage to judge the models created and audit the inference outcomes, and (v) **reproducibility**—principle to trace all methodology including the machine learning pipeline code, input arguments, versioned model and historic/input data.

The well-known big data ecosystem called Hadoop⁸ was the platform in which the purposed system was implemented. For storage, the column-oriented database HBase⁹ and a designed data schema are applied to take advantage of the scalable database: ASSET, ASSET-GEOINDEX, MODEL-CONF, SESSION, MEASUREMENT, and FORECAST. The first three tables do not exhibit characteristics for the need of a big data storage; HBase was used to simplify the architecture and keep with a unique database instance. Indeed, the other three tables store a large amount of data volume, in which the last two are appended daily by data pipelines engineering. For computing processing, YARN is used as the platform to handle the parallelism during the individual calibration and forecasting of models which runs as a YARN application.

⁸ Apache Hadoop Project at <https://hadoop.apache.org/>.

⁹ Apache HBase Project at <https://hbase.apache.org/>.

Table A.10

GAMLFL-SL results calculated over the time span from 2016 and 2019 using cross-validation of timeseries with a fixed 3-year window for training and the next 1 year (1 fold), 2 weeks (26 folds), 1 week (52 folds), or 1 day (365 folds) of testing data. MASE was calculated with m steps equal to 52 weeks, 2 weeks, 1 week and 1 day, respectively. The minimum, maximum, mean, and the first, second (median) and third quartiles are calculated for all metrics on each cross-validation fold.

	Model	1 year	2 weeks						1 week						1 day					
		Median	Min	1Q	Median	3Q	Max	Mean	Min	1Q	Median	3Q	Max	Mean	Min	1Q	Median	3Q	Max	Mean
MAE (MW)	GLMLF-B	235.25	99.33	133.99	185.46	246.94	606.27	217.75	76.19	131.78	174.26	257.90	760.52	214.64	48.77	113.11	147.17	238.50	1916.92	210.46
	GAMLF-SL-M1	226.86	96.10	136.54	192.13	230.32	241.92	201.63	83.48	128.19	170.11	242.41	489.81	199.88	51.85	116.67	146.84	217.61	1581.24	193.66
	GAMLF-SL-M2	140.77	64.17	91.92	105.84	173.89	317.19	137.89	60.96	86.36	110.16	155.70	390.15	137.67	25.32	71.25	97.28	147.37	974.48	136.63
	GAMLF-SL-M3	126.97	60.48	90.41	115.65	139.73	263.02	123.91	57.13	83.67	110.15	135.88	329.43	123.62	25.38	69.66	90.01	135.80	858.93	122.69
MAPE (%)	GLMLF-B	4.23	1.75	2.44	3.25	4.74	11.48	3.87	1.40	2.38	2.84	4.85	15.18	3.82	0.83	1.90	2.62	4.07	41.53	3.75
	GAMLF-SL-M1	4.09	1.73	2.47	3.23	4.13	7.90	3.59	1.50	2.23	2.93	4.14	9.77	3.56	0.88	2.02	2.62	3.71	34.26	3.46
	GAMLF-SL-M2	2.53	1.15	1.57	1.91	3.29	6.06	2.47	1.11	1.54	1.82	2.56	7.66	2.46	0.44	1.24	1.66	2.45	21.49	2.44
	GAMLF-SL-M3	2.26	1.07	1.58	1.93	2.53	4.83	2.20	1.02	1.49	1.86	2.23	6.57	2.19	0.43	1.19	1.60	2.30	16.74	2.18
RMSE (MW)	GLMLF-B	353.12	123.09	170.61	259.29	390.05	814.17	296.99	95.06	166.22	215.53	386.01	979.36	283.73	60.54	138.97	181.33	287.26	2049.36	249.74
	GAMLF-SL-M1	322.65	122.33	175.85	254.70	336.20	585.08	275.59	108.50	160.03	218.70	354.80	738.20	264.56	66.88	142.60	178.94	255.63	1747.64	233.66
	GAMLF-SL-M2	229.16	84.06	118.43	137.38	287.37	461.68	198.11	76.81	105.26	140.27	196.57	549.17	189.97	35.52	88.37	116.09	175.75	1093.63	165.63
	GAMLF-SL-M3	191.08	76.31	117.85	149.17	190.02	374.88	172.28	74.28	103.50	143.08	174.43	437.58	165.82	32.88	85.90	112.40	160.49	982.90	148.25
NRMSE (%)	GLMLF-B	6.14	2.20	3.05	4.13	6.58	14.43	5.17	1.69	2.99	3.71	6.42	18.46	4.96	1.02	2.40	3.20	4.78	43.29	4.44
	GAMLF-SL-M1	5.61	2.18	3.14	4.09	6.17	10.37	4.81	1.98	2.84	3.61	5.88	12.71	4.64	1.14	2.48	3.14	4.46	36.91	4.16
	GAMLF-SL-M2	3.99	1.47	2.09	2.41	5.27	8.20	3.48	1.40	1.87	2.30	3.43	9.48	3.35	0.61	1.50	2.05	2.97	23.28	2.97
	GAMLF-SL-M3	3.32	1.33	2.09	2.56	3.49	6.66	3.01	1.32	1.83	2.37	3.07	8.32	2.91	0.56	1.50	1.95	2.83	19.39	2.64
R ²	GLMLF-B	0.888	0.597	0.877	0.963	0.974	0.985	0.918	0.614	0.943	0.971	0.980	0.988	0.929	0.342	0.965	0.980	0.988	0.997	0.958
	GAMLF-SL-M1	0.909	0.723	0.870	0.963	0.975	0.985	0.927	0.637	0.946	0.971	0.978	0.988	0.933	0.312	0.964	0.977	0.986	0.996	0.957
	GAMLF-SL-M2	0.948	0.763	0.903	0.980	0.986	0.993	0.944	0.696	0.964	0.983	0.988	0.993	0.946	0.189	0.983	0.991	0.995	0.998	0.967
	GAMLF-SL-M3	0.965	0.851	0.964	0.978	0.983	0.994	0.963	0.717	0.966	0.981	0.989	0.994	0.964	0.444	0.984	0.991	0.995	0.998	0.977
MASE	GLMLF-B	0.519	0.323	0.486	0.661	0.902	1.916	0.742	0.220	0.643	0.850	1.263	3.212	0.970	0.063	0.263	0.633	1.718	9.193	1.207
	GAMLF-SL-M1	0.501	0.316	0.485	0.623	0.870	1.274	0.699	0.241	0.619	0.833	1.197	2.188	0.907	0.073	0.254	0.621	1.518	9.300	1.142
	GAMLF-SL-M2	0.311	0.213	0.347	0.425	0.590	0.962	0.468	0.264	0.427	0.561	0.727	1.212	0.603	0.056	0.171	0.410	1.001	4.298	0.686
	GAMLF-SL-M3	0.280	0.170	0.316	0.399	0.514	0.828	0.433	0.232	0.344	0.539	0.697	1.620	0.568	0.063	0.159	0.368	0.914	4.894	0.653

The application submits a new YARN job submission to `ApplicationManager` which handles the negotiation of the first container. Consequently, the `ApplicationMaster` coordinates the logical plan of the calibration or forecasting job by requesting resources from `ResourceManager`, generating a physical plan from the resources it receives, and coordinating the execution of the plan around faults.

The system is deployed in a Hadoop cluster with 22 servers. HBase, YARN, HDFS, and services related to Hadoop ecosystem are deployed in the same cluster, tweaking the CPU and memory shared by the services in accordance with Hadoop best practices. The resulting live daily forecasting system called PREDIS Portuguese acronym for DIStributed PREdiction is recognized as an application prescribed and maintained by the DSO. This is possible by a distributed system architecture which copes with the challenge of capacity and scalability for the storage and computing of hundreds of thousands of time series and artifacts.

One of the challenges is the applicability of individual forecasting models, which must be run daily to infer the next 24 h. Specifically, the head end system for advanced metering infrastructure collects the updated power load time series throughout the day from all secondary substations and ends in the first hours of dawn. After updating the modeled data in Hbase, the system has a few hours to individually infer all 100,000 time series. Indeed, for the forecasting process of 100,000 individual models and considering 100 vcores available in the Hadoop system, the process takes 5h42', which corresponds to 23d18h16' vcores time, that is, the accumulated time the process would take sequentially. Note that the Hadoop cluster with 22 servers is configured up to approximately 550 vcores available to YARN in which the process runs. As the process scales linearly at least in the range of 100–550 vcores, the inference computing time is parallelizable enough to have the outputs in useful time to decision-marking and operations.

4. Conclusions

This study proposes a systematic approach to forecast the power load from system level to low voltage level. At this particular level, DSOs have rolled out smart meters in secondary substations over the past few years and have been collecting disaggregated measurements

since then. While the sensorization of the power grid advances, the availability of data combined with the desire to improve maintenance, operations, and predictability of the grid, as well as the eagerness to optimize and unlock business models through technology, increases the need for new tools and methods, such as the possibility to forecast the power load at all secondary substations.

This study considered not only the performance of the models, as many other research have been focused on, but also the applicability, interpretability, and reproducibility of the method/model considering the context of a DSO in a highly regulated business per se in a Europe that seeks to regulate AI across several applications, particularly the models that act or support decisions over critical infrastructure.

An initial benchmark model described by Tao Hong and his research group was with improved models enhanced by introducing new synthetic explanatory variables and a different regression technique, GAM. The errors were reduced by 42%–47% compared with the benchmark model, preserving the interpretability of outcomes. Additionally, a simple ensemble method was assessed how easily improves the accuracy for specific periods whose modeling is particularly difficult using GAM, like weekends, summer leave, or public holidays. The method was applied to national power load and, for the first time, to all 100,000 secondary substations that integrate the Portugal power grid. The approach was used to build a daily forecasting *live*¹⁰ system called PREDIS whose results are used to anticipate load peaks and network constraints.

CRedit authorship contribution statement

Marco G. Pinheiro: Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Sara C. Madeira:** Supervision. **Alexandre P. Francisco:** Writing – review & editing, Supervision.

¹⁰ Deployed in production environment.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The author Marco Pinheiro declares that he is a EDP employee, a private electric utility company with interests on the subject. Part of the work reported in this paper was integrated in an internal project called PREDIS. EDP sponsors financially part of the tuition fees of the author's doctoral studies. The authors Sara Madeira and Alexandre Francisco declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

Appendix. Accuracy of models

See Table A.10

References

- Markard J. The next phase of the energy transition and its implications for research and policy. *Nat Energy* 2018;3(8):628–33.
- Nikobakht A, Aghaei J, Shafie-Khah M, Catalao JP. Assessing increased flexibility of energy storage and demand response to accommodate a high penetration of renewable energy sources. *IEEE Trans Sustain Energy* 2018;10(2):659–69.
- Lopes JP, Almeida PMR, Silva AM, Soares FJ. Smart charging strategies for electric vehicles: Enhancing grid performance and maximizing the use of variable renewable energy resources. 2009.
- Oliveira AM, Beswick RR, Yan Y. A green hydrogen economy for a renewable energy society. *Curr Opin Chem Eng* 2021;33:100701.
- Hong T. Short term electric load forecasting [Ph.D. thesis], North Carolina State University; 2010, URL <http://repository.lib.ncsu.edu/ir/handle/1840.16/6457>.
- Ahmed A, Khalid M. A review on the selected applications of forecasting models in renewable power systems. *Renew Sustain Energy Rev* 2019;100:9–21. <http://dx.doi.org/10.1016/j.rser.2018.09.046>, URL <https://www.sciencedirect.com/science/article/pii/S1364032118306932>.
- Hernandez L, Baladron C, Aguiar JM, Carro B, Sanchez-Esguevillas AJ, Lloret J, et al. A survey on electric power demand forecasting: Future trends in smart grids, microgrids and smart buildings. *IEEE Commun Surv Tutor* 2014;16(3):1460–95. <http://dx.doi.org/10.1109/SURV.2014.032014.00094>.
- Mujeeb S, Javaid N, Javaid S. Data analytics for price forecasting in smart grids: A survey. In: 2018 IEEE 21st international multi-topic conference. 2018, p. 1–10. <http://dx.doi.org/10.1109/INMIC.2018.8595571>.
- Willis HL. Power distribution planning reference book, second edition, revised and expanded. CRC Press; 2004.
- Amarasinghe K, Marino DL, Manic M. Deep neural networks for energy load forecasting. In: 2017 IEEE 26th international symposium on industrial electronics. ISIE, 2017, p. 1483–8. <http://dx.doi.org/10.1109/ISIE.2017.8001465>.
- Jabir HJ, Teh J, Ishak D, Abunima H. Impacts of demand-side management on electrical power systems: A review. *Energies* 2018;11(5). <http://dx.doi.org/10.3390/en11051050>, URL <https://www.mdpi.com/1996-1073/11/5/1050>.
- Hobbs BF, Jitrapakulsarn S, Konda S, Chankong V, Loparo KA, Maratukulam DJ. Analysis of the value for unit commitment of improved load forecasts. *IEEE Trans Power Syst* 1999;14(4):1342–8.
- Stimmel CL. Big data analytics strategies for the smart grid. Auerbach Publications; 2019.
- Dudek G. Pattern similarity-based methods for short-term load forecasting—Part 1: Principles. *Appl Soft Comput* 2015;37:277–87.
- Petropoulos F, Apiletti D, Assimakopoulos V, Babai MZ, Barrow DK, Taieb SB, et al. Forecasting: Theory and practice. 2021, [arXiv:2012.03854](https://arxiv.org/abs/2012.03854).
- Matthewman P, Nicholson H. Techniques for load prediction in the electricity-supply industry. *Proc Inst Electr Eng* 1968;115(10):1451. <http://dx.doi.org/10.1049/piee.1968.0258>.
- Gross G, Galiana FD. Short-term load forecasting. *Proc IEEE* 1987;75(12):1558–73. <http://dx.doi.org/10.1109/proc.1987.13927>.
- Moghran I, Rahman S. Analysis and evaluation of five short-term load forecasting techniques. *IEEE Trans Power Syst* 1989;4(4):1484–91.
- Alfares HK, Nazeeruddin M. Electric load forecasting: Literature survey and classification of methods. *Internat J Systems Sci* 2002;33(1):23–34.
- Hahn H, Meyer-Nieberg S, Pickl S. Electric load forecasting methods: Tools for decision making. *European J Oper Res* 2009;199(3):902–7. <http://dx.doi.org/10.1016/j.ejor.2009.01.062>.
- Suganthi L, Samuel AA. Energy models for demand forecasting — A review. *Renew Sustain Energy Rev* 2012;16(2):1223–40. <http://dx.doi.org/10.1016/j.rser.2011.08.014>.
- Hong T, Fan S. Probabilistic electric load forecasting: A tutorial review. *Int J Forecast* 2016;32(3):914–38.
- Kuster C, Rezgui Y, Mourshed M. Electrical load forecasting models: A critical systematic review. *Sustainable Cities Soc* 2017;35:257–70.
- Fallah SN, Ganjkhani M, Shamsirband S, Chau K-w. Computational intelligence on short-term load forecasting: A methodological overview. *Energies* 2019;12(3):393.
- Nti IK, Teimeh M, Nyarko-Boateng O, Adekoya AF. Electricity load forecasting: A systematic review. *J Electr Syst Inf Technol* 2020;7(1):1–19.
- Hong T, Pinson P, Wang Y, Weron R, Yang D, Zareipour H. Energy forecasting: A review and outlook. *IEEE Open Access J Power Energy* 2020;7:376–88. <http://dx.doi.org/10.1109/OAJPE.2020.3029979>, URL <https://ieeexplore.ieee.org/abstract/document/9218967>.
- Pappas SS, Ekonomou L, Karamousantas DC, Chatzarakis G, Katsikas S, Liatsis P. Electricity demand loads modeling using AutoRegressive moving average (ARMA) models. *Energy* 2008;33(9):1353–60.
- Lee Y-S, Tong L-I. Forecasting time series using a methodology based on autoregressive integrated moving average and genetic programming. *Knowl-Based Syst* 2011;24(1):66–72.
- Chakhchoukh Y, Panciatici P, Mili L. Electric load forecasting based on statistical robust methods. *IEEE Trans Power Syst* 2010;26(3):982–91.
- Sigauke C, Chikobvu D. Peak electricity demand forecasting using time series regression models: An application to South African data. *J Stat Manag Syst* 2016;19(4):567–86.
- Caro E, Juan J, Cara J. Periodically correlated models for short-term electricity load forecasting. *Appl Math Comput* 2020;364:124642.
- Hong T, Wang P, Willis HL. A naïve multiple linear regression benchmark for short term load forecasting. In: 2011 IEEE power and energy society general meeting. IEEE; 2011, p. 1–6.
- Pierrot A, Goude Y. Short-term electricity load forecasting with generalized additive models. *Proc ISAP Power* 2011;2011.
- Cho H, Goude Y, Brossat X, Yao Q. Modelling and forecasting daily electricity load via curve linear regression. In: Modeling and stochastic learning for forecasting in high dimensions. Springer; 2015, p. 35–54.
- Amato U, Antoniadis A, De Feis I, Goude Y, Lagache A. Forecasting high resolution electricity demand data with additive models including smooth and jagged components. *Int J Forecast* 2021;37(1):171–85.
- Lebotsa ME, Sigauke C, Bere A, Fildes R, Boylan JE. Short term electricity demand forecasting using partially linear additive quantile regression with an application to the unit commitment problem. *Appl Energy* 2018;222:104–18.
- Reis M, Garcia A, Bessa RJ. A scalable load forecasting system for low voltage grids. In: 2017 IEEE Manchester powertech. IEEE; 2017, p. 1–6.
- Viana J, Bessa RJ, Sousa J. Load forecasting benchmark for smart meter data. In: 2019 IEEE Milan powertech. IEEE; 2019, p. 1–6.
- Khotanzad A, Afkhami-Rohani R, Lu T-L, Abaye A, Davis M, Maratukulam DJ. ANNSTLF-a neural-network-based electric load forecasting system. *IEEE Trans Neural Netw* 1997;8(4):835–46.
- Kumar S, Mishra S, Gupta S. Short term load forecasting using ANN and multiple linear regression. In: 2016 Second international conference on computational intelligence & communication technology. IEEE; 2016, p. 184–6.
- Khawaja AS, Anpalagan A, Naeem M, Venkatesh B. Joint bagged-boosted artificial neural networks: Using ensemble machine learning to improve short-term electricity load forecasting. *Electr Power Syst Res* 2020;179:106080.
- Choi H, Ryu S, Kim H. Short-term load forecasting based on ResNet and LSTM. In: 2018 IEEE international conference on communications, control, and computing technologies for smart grids. IEEE; 2018, p. 1–6.
- Bouktif S, Fiaz A, Ouni A, Serhani MA. Optimal deep learning lstm model for electric load forecasting using feature selection and genetic algorithm: Comparison with machine learning approaches. *Energies* 2018;11(7):1636.
- Lim B, Arık SÖ, Loeff N, Pfister T. Temporal fusion transformers for interpretable multi-horizon time series forecasting. *Int J Forecast* 2021.
- Chen B-J, Chang M-W, et al. Load forecasting using support vector machines: A study on EUNITE competition 2001. *IEEE Trans Power Syst* 2004;19(4):1821–30.
- Niu D, Wang Y, Wu DD. Power load forecasting using support vector machine and ant colony optimization. *Expert Syst Appl* 2010;37(3):2531–9.
- Khosravi A, Nahavandi S, Creighton D, Srinivasan D. Interval type-2 fuzzy logic systems for load forecasting: A comparative study. *IEEE Trans Power Syst* 2012;27(3):1274–82.
- Hong T, Wang P. Fuzzy interaction regression for short term load forecasting. *Fuzzy Optim Decis Mak* 2014;13(1):91–103.
- Singla MK, Hans S. Load forecasting using fuzzy logic tool box. *Glob Res Dev J Eng* 2018;38:12–9.
- Bogomolov A, Lepri B, Larcher R, Antonelli F, Pianesi F, Pentland A. Energy consumption prediction using people dynamics derived from cellular network data. *EPJ Data Sci* 2016;5:1–15.
- Wang J, Zhu S, Zhang W, Lu H. Combined modeling for electric load forecasting with adaptive particle swarm optimization. *Energy* 2010;35(4):1671–8.

- [52] Qiang S, Pu Y. Short-term power load forecasting based on support vector machine and particle swarm optimization. *J Algorithms Comput Technol* 2018;13:1748301818797061.
- [53] Li Y, Zhang S, Hu R, Lu N. A meta-learning based distribution system load forecasting model selection framework. *Appl Energy* 2021;294:116991.
- [54] Lee E, Rhee W. Individualized short-term electric load forecasting with deep neural network based transfer learning and meta learning. *IEEE Access* 2021;9:15413–25.
- [55] Xiao X, Mo H, Zhang Y, Shan G. Meta-ANN—A dynamic artificial neural network refined by meta-learning for short-term load forecasting. *Energy* 2022;246:123418.
- [56] Kong W, Dong ZY, Jia Y, Hill DJ, Xu Y, Zhang Y. Short-term residential load forecasting based on LSTM recurrent neural network. *IEEE Trans Smart Grid* 2019;10(1):841–51. <http://dx.doi.org/10.1109/TSG.2017.2753802>.
- [57] Raza MQ, Khosravi A. A review on artificial intelligence based load demand forecasting techniques for smart grid and buildings. *Renew Sustain Energy Rev* 2015;50:1352–72. <http://dx.doi.org/10.1016/j.rser.2015.04.065>.
- [58] Kong X, Li C, Wang C, Zhang Y, Zhang J. Short-term electrical load forecasting based on error correction using dynamic mode decomposition. *Appl Energy* 2020;261:114368.
- [59] Haben S, Arora S, Giasemidis G, Voss M, Vukadinović Greetham D. Review of low voltage load forecasting: Methods, applications, and recommendations. *Appl Energy* 2021;304:117798. <http://dx.doi.org/10.1016/j.apenergy.2021.117798>, URL <https://www.sciencedirect.com/science/article/pii/S0306261921011326>.
- [60] Li R, Gu C, Li F, Shaddick G, Dale M. Development of low voltage network templates—Part I: Substation clustering and classification. *IEEE Trans Power Syst* 2014;30(6):3036–44.
- [61] Li R, Gu C, Li F, Shaddick G, Dale M. Development of low voltage network templates—Part II: Peak load estimation by clusterwise regression. *IEEE Trans Power Syst* 2014;30(6):3045–52.
- [62] Taborda D, De Matos AA, Geria J, Fortunato S, Silva J. Secondary substations smart metering campaign. *CIREN-Open Access Proc J* 2017;2017(1):2893–6.
- [63] Bahmanyar A, Jamali S, Estebars A, Pons E, Bompard E, Patti E, et al. Emerging smart meters in electrical distribution systems: Opportunities and challenges. In: 2016 24th Iranian conference on electrical engineering. IEEE; 2016, p. 1082–7.
- [64] Löf N, Pikkarainen M, Repo S, Järventausta P. Utilizing smart meters in LV network management. In: CIREN seminar, vol. 2011, 2011, p. 21st.
- [65] Hastie TJ, Tibshirani RJ. Generalized additive models. *Statist Sci* 1986;297–310.
- [66] Hastie TJ, Tibshirani RJ. Generalized additive models, Vol. 43. CRC Press; 1990.
- [67] Wood SN. Generalized additive models: An introduction with R. Chapman & Hall; 2006.
- [68] Wood SN. Mgcov: Mixed GAM computation vehicle with automatic smoothness estimation. 2021, R package version 1.8-38.
- [69] Natekin A, Knoll A. Gradient boosting machines, a tutorial. *Front Neurobot* 2013;7:21.
- [70] Friedman J, Hastie T, Tibshirani R. Additive logistic regression: A statistical view of boosting. *Ann Statist* 2000;28(2):337–407.
- [71] Friedman JH. Greedy function approximation: A gradient boosting machine. *Ann Statist* 2001;1189–232.
- [72] Abbasi RA, Javaid N, Ghuman MNJ, Khan ZA, Ur Rehman S, et al. Short term load forecasting using XGBoost. In: Workshops of the international conference on advanced information networking and applications. Springer; 2019, p. 1120–31.
- [73] Liao X, Cao N, Li M, Kang X. Research on short-term load forecasting using XGBoost based on similar days. In: 2019 International conference on intelligent transportation, big data & smart city. IEEE; 2019, p. 675–8.
- [74] Wang Y, Sun S, Chen X, Zeng X, Kong Y, Chen J, et al. Short-term load forecasting of industrial customers based on SVM and XGBoost. *Int J Electr Power Energy Syst* 2021;129:106830.
- [75] Littlestone N, Warmuth MK. The weighted majority algorithm. *Inform and Comput* 1994;108(2):212–61. <http://dx.doi.org/10.1016/B978-0-08-094829-4.50035-0>, URL <https://users.soe.ucsc.edu/~manfred/pubs/J24.pdf>.
- [76] Kim B, Khanna R, Koyejo OO. Examples are not enough, learn to criticize! Criticism for interpretability. *Adv Neural Inf Process Syst* 2016;29.
- [77] Hyndman RJ, Koehler AB. Another look at measures of forecast accuracy. *Int J Forecast* 2006;22(4):679–88.
- [78] Haben S, Ward J, Greetham DV, Singleton C, Grindrod P. A new error measure for forecasts of household-level, high resolution electrical energy consumption. *Int J Forecast* 2014;30(2):246–56.
- [79] Hyndman RJ, Athanasopoulos G. Forecasting: Principles and practice. OTexts; 2018.
- [80] Gunning D, Stefik M, Choi J, Miller T, Stumpf S, Yang G-Z. XAI—Explainable artificial intelligence. *Science Robotics* 2019;4(37):eaay7120.
- [81] Došilović FK, Brčić M, Hlupić N. Explainable artificial intelligence: A survey. In: 2018 41st International convention on information and communication technology, electronics and microelectronics. IEEE; 2018, p. 0210–5.
- [82] Angelov PP, Soares EA, Jiang R, Arnold NI, Atkinson PM. Explainable artificial intelligence: An analytical review. *Wiley Interdiscipl Rev: Data Min Knowl Discov* 2021;11(5):e1424.
- [83] Molnar C, König G, Herbringer J, Freiesleben T, Dandl S, Scholbeck CA, et al. General pitfalls of model-agnostic interpretation methods for machine learning models. In: International workshop on extending explainable AI beyond deep models and classifiers. Springer; 2022, p. 39–68.
- [84] Gonçalves R, Louro M, Paulo A, Ferreira P, Pinheiro M, Pedro M, et al. Forecasted chronological power flow for enabling timely dynamic tariff activation. In: CIREN 2019 conference. AIM; 2019.